

Agent-based modelling for network-wide COVID-19 mitigation strategies for TfL

Abstract: The aim of this project is to provide a method for analysing disease spread in urban rail networks so that policy decisions can be made more effectively. The SARS-CoV-2 (COVID-19) pandemic has demonstrated the scale to which disease can disrupt modern society. With global pandemics predicted to become more common, questions are presented as to how to analyse disease spread in public transport environments and how to assess the effectiveness of different mitigation methods.

A toolkit approach was developed for this purpose. An agent-based model of disease transmission in an urban rail station was created for analysing the perceived risk that a station poses to passengers. This is used to compare policy options so that an optimal risk mitigation strategy can be implemented. Analysis of network data through K-means clustering is used to discover similar stations on the network to transfer this policy to. Additionally, the agent-based model is used to assign busyness-based risk values to station facilities and, combined with network usage data, to assign initial risk values to all network stations. This is employed to quickly highlight which stations on the network pose the highest risk. These values are also used to route network passengers to avoid risk via utilisation of a dynamically applied Dijkstra's algorithm.

This project presents a proof of concept for this approach, although performs analysis on the London Underground network for examples. Due to the time constraints imposed and limited data available, no genuine suggestion of optimal policy is presented.

Analysis of a high-risk station resulted in an 80% decrease in the probability of infection through policy implementation. Data processing techniques show that this finding is transferable to 15 other similar stations in the network, hence indicating a possible decrease of approximately 2.5% in those infected across the 15 stations, corresponding to around 12200 people. It is also suggested that by implementing routing to avoid risk as policy to decrease risk of infection there is the potential to decrease passenger risk of infection by up to 50% relative to travelling via the fastest route between origin and destination.

It is demonstrated that this approach can be used by those in network management roles to effectively implement mitigation policy and save lives as a result.



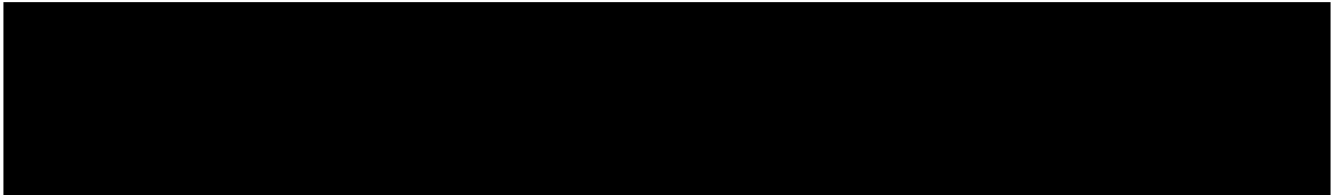
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Declaration

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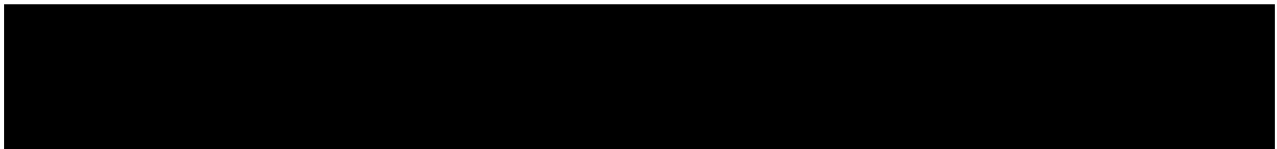
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Work Allocation

Section	Student Contribution			
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Abstract	X(10%)	X(10%)	X(70%)	X(10%)
1	X(70%)	X(10%)	X(10%)	X(10%)
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12	X(10%)	X(10%)	X(70%)	X(10%)

We confirm that the information on this page accurately describes our individual contributions to the project.



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1 Introduction

The SARS-CoV-2 (COVID-19) pandemic has shown the large-scale disruption to our society that such a disease can have. Its fast spread around cities, countries and the globe is easily attributed to the combined factors of the disease's high infectivity and the connectedness of modern society. Public transportation can be highlighted as a vector of this urban transmission [1] as it offers a culmination of feared dangerous factors: a large and diverse ridership, poor ventilation, enclosed environments and numerous touch points with often lax cleaning.

Due to these conditions, many public transportation systems faced suspension and implementation of heavy risk mitigation policy under the COVID-19 pandemic [2]. As society continues to become more connected, it is expected that global pandemics will only become more likely. The characteristics of future diseases are unknown thus it is difficult to plan ahead, but, preparing tools to analyse disease spread in preparation would be advantageous [3].

In a pandemic the ability to experiment and collect data is limited. While retrospective case studies can be performed, these can rarely be used to provide enough insight to make the most effective policy decisions [4]. It is in cases such as these where modelling is implemented. Modelling is the process of creating an abstraction of the real world so that experimentation can be performed in a risk-free environment [5].

Agent-Based Modelling (ABM) is a type of simulation modelling. ABM is useful in situations where the dynamics of individuals (not necessarily people) and the interactions between them are known but the behaviour of the whole system is not [6]. ABM has often been applied to the study of pandemics, including the COVID-19 pandemic, at a range of scales [7–9].

Its potential use for study of transmission on urban rail systems, with the aim of improving policy implementation efficiency, has been hypothesised. This report presents the findings of an investigation into its efficacy in this scenario, providing a proof of concept for a toolkit to assist in decision making.

1.1 Background

To aid in understanding the motives of system managers facing this situation, a real network will be used for analysis. This is chosen to be the London Underground (LU) urban rail system. Transport for London (TfL), which operates the LU along with a large proportion of public transportation in London, faced difficult decisions in the recent pandemic [2].

To ensure that the approach taken will result in the optimal method, the key stakeholders in this situation were considered. These were identified to be:

- Network Executives - Control and ensure commercial viability of the network.
- Network Managers - Manage the function and operation of the network.
- Station Managers - Manage an individual station.
- Station Workers - Cleaners, passenger assistants and security guards who work at a given station.
- Network Passengers - The customers of the network.
- Modellers - Personnel who perform the suggested analysis.

The method must rely on historical data as data collection would take up vital time. It must also be functional for a wide range of diseases for which a train station would be a major vector for transmission.

1.2 Aims

The aim of this project is to provide a method and toolkit for analysing diseases in urban rail networks so that policy decisions can be effectively made. Considering the needs of the stakeholders outlined in the previous section, the following objectives were set:

1. Provide a method to quickly analyse network risk so that focus can be placed on high risk stations.
2. Develop a model for disease transmission with focus on that of the COVID-19 virus.
3. Investigate integration of a disease transmission model with a generalised station modelling approach.
4. Utilise an integrated station disease transmission model to investigate the effect of COVID-19 risk mitigation policy at urban rail stations.
5. Investigate how policy can be transferred to additional stations without development of an additional model.
6. Evaluate what actions can be taken by network passengers to minimise risk.

To complete these objectives a selection of components will be required. These will be covered in a logical manner based on dependencies, rather than in order of objective. Section 10 will provide an example of how these components assemble to create a toolkit which can be used to achieve the objectives.

2 Literature Review

This project involves the culmination of aspects from multiple disciplines and areas of research. Therefore, before beginning work on any of the objectives, extensive research into the main fields of research utilised in this study was conducted to gain insight into the state-of-the-art. The following section outlines the relevant information found and how this influenced decisions on processes used.

2.1 Diseases and Transmission Routes

Modelling disease transmission is a well-studied and broad field due to the large range of communicable diseases [10]. The possible mechanisms behind disease transmission vary greatly with the disease but can be categorised into direct and in-direct transmission.

Direct transmission concerns methods via direct physical contact and droplet transmission. The latter is transmission via large, short-range aerosols often produced by breathing or coughing and sneezing. Indirect transmission consists of infection via vehicles of infection (eg. shared surfaces), disease vectors (mosquitoes, fleas, etc.) and long-range aerosol transmission. Long-range aerosol transmission is similar to droplet transmission although particles are smaller and remain in the air for a longer time.

The relative importance of each of these transmission routes varies greatly with the disease considered. Whilst there are a wide range of disease types which hold the potential to be the next major pandemic, this report focuses on the last major global pandemic, COVID-19. COVID-19 is a respiratory disease; as such the major transmission routes posed are considered: short-range droplet, long-range aerosol, and in-direct contact via fomites [11, 12]. These are shown in Figure 1. The key transmission routes are well known but the relative contribution of each is disputed.

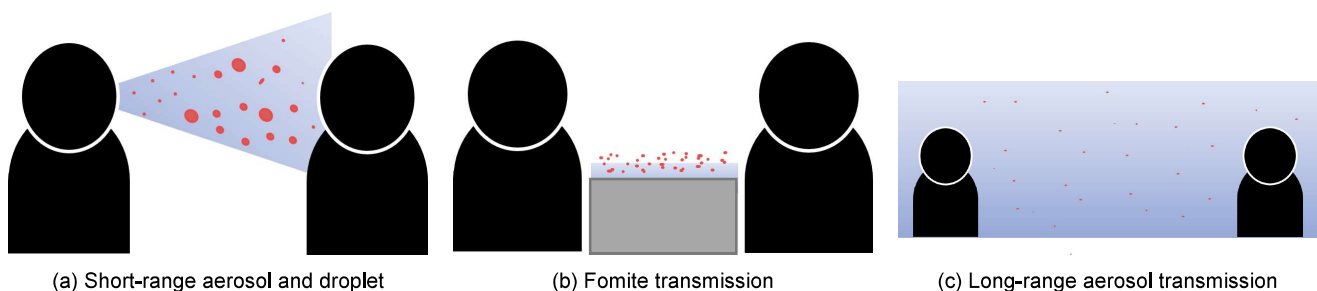


Figure 1: Key transmission routes for COVID-19

2.2 Disease Modelling Techniques

Several approaches can be used to model the transmission of disease through a population. These vary in scale, application and approach to risk quantification.

One technique is compartmental mechanistic modelling [13]. This sees the modelled population split into compartments that reflect the possible stages of infection. Transitions between compartments are determined by a system of differential equations, the coefficients of which are derived from case data [13, 14]. The simplest form of compartmental model is an SI (Susceptible - Infected) model, but the complexity in the compartments considered can vary dependent on both the disease in question and the level of abstraction modelled. Stochastic compartmental models are an alternative to the previous mechanistic approach [15]. Both of these approaches are more applicable at larger scales than considered in this study as they fail to consider the complex disease transmission dynamics present [16].

Agent-based modelling has been proven to be a suitable approach for a range of different diseases [17, 18], on a range of scales [7–9], and is applicable in a wide range of modelling scenarios [19]. Zhou et al. applies ABM to the modelling of COVID at a community scale within a university campus [7]. Similar compartmental states to previous work are again used, but the transitions between states are dependent on interactions between agents. This took into account disease intervention strategies (face masks and vaccination) as well as physical disease transmission parameters, including contact time and distance. Factors were applied to a base calibration parameter for each interaction when calculating

probability of infection. Grignard et al. similarly applies ABM for the study of COVID-19 transmission at a university, however, includes specific disease transmission dynamics for transmission routes including droplets, fomites and aerosols via viral quantification [20].

ABMs are of particular use in situations where interactions are an important factor the modelled system [4], which is the case for infectious disease. It allows for the consideration of human behaviour as well the characteristics of the population [4]. Therefore, ABM will be selected as the modelling approach taken.

2.3 Assessing Risk to the Individual

With ABM selected, the transitions between agent states must be investigated with respect to transmission routes. Several approaches seek to quantify the total contaminant received by an individual. This has been performed over a range of transmission routes and is often applied at scales comparable to that considered in this report.

Computational Fluid Dynamics (CFD) is a commonly used approach to assess the risk of aerosolised particles and droplets [21–24]. This has been applied to a wide range of situations and tends to be applied to smaller scales, ranging from modelling risk within elevators to hospital wards [21–23]. Whilst CFD is a powerful tool for assessing the relative risk presented by aerosols in the spacial domain, sources that solely use CFD cannot adequately quantify the risk posed to the individual and often use aerosol density as a proxy for risk [21, 24]. CFD also requires topographical details [22] which are not publicly accessible for Underground stations. The larger stations within the LU network would also present significant computational cost if fully modelled and so complete CFD modelling cannot be considered a viable option in this case.

Cortallessa et al. utilises viral quantification combined with the exponential dose response model to determine infection risk between two people in a conversational setting [25]. CFD is used to estimate the absorbed aerosols of a subject at varying transmission distances and time. This is then used in combination with the viral load of a COVID-19 infected person to quantify the total viral quantity received by the susceptible.

Viral quantification and the use of the dose-response model is not limited to the transmission route of droplets and aerosols. Nicas and Best utilise the exponential dose response model to present a pioneering framework to assess the hand-to-face contact rate and its impact on predicting the likelihood of a respiratory tract infection [26]. By assessing the contact rate of a hand to face of ten subjects in an office environment a model was created to assess the total microbe transferred to target membranes and by extension the dose received. Several papers follow this approach but differences emerge when considering the behaviour of the people is estimated through the use of stochastics or through mechanistic implementation [27, 28].

Alternatively to the dose response model, the Wells Riley model assesses the risk posed by aerosols. This calculates an infection probability based on a range of inputs regarding the infectious space and the people within. It assumes a well mixed room and applies a blanket probability of infection to all people within the room, failing to account for position and flows within the room. Therefore although the Wells Riley method has seen wide spread application [29, 30], it is deemed unsuitable.

The dose response model was selected due to its wide applicability to a large number of diseases and provides a more parameterisable approach than COVID-19 specific models. In addition to this, the dose response model allows for dose contributions from different transmission routes to be integrated with ease in to the infection probability calculation. The dose response model also allows for the probability of infection to be individualised to each agent rather than the approach in the Wells Riley model [29].

2.4 Modelling Pedestrian Movement

Attention now shifts to how the infected and uninfected individuals interact in the physical environment. Alodhaibi et al. [31] present a DES approach to modelling an airport terminal. Discrete Event Simulation (DES) which considers systems as a set of processes occurring sequentially or in parallel [5]; key features of an airport (check-in lines, security screening, ect.) can be described this way. However, this does not consider the infrastructure of the airport and the constraints this imposes.

Cheng et al. [32] analyses an airport although used an agent-based approach. While the services around the airport remained discrete-event modelled, movement between these services was

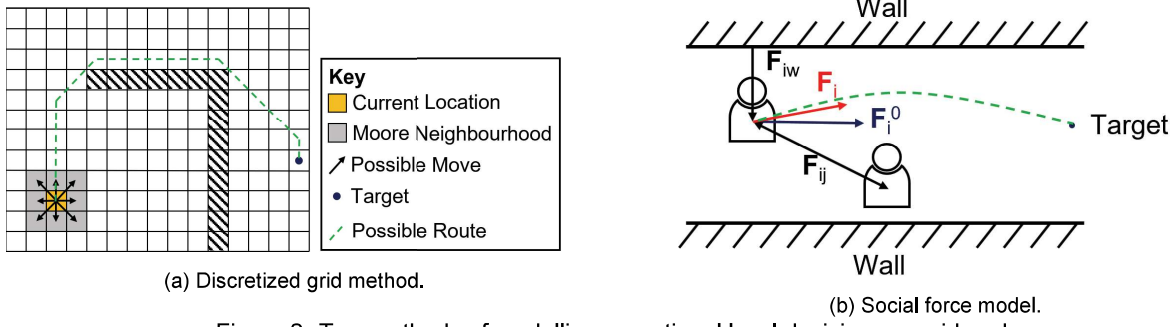


Figure 2: Two methods of modelling operational level decisions considered.

performed by autonomous agents. This allows for the environment to be considered and possibly constrain the system. This low-abstraction process is known as microsimulation [33] and, as the aim is to analyse disease transmission at the microscale level, this will be the approach taken.

Hoogendoorn [34] breaks pedestrian movement down into three levels of decisions. The highest level is strategic which incorporates where the pedestrian is starting their journey, their destination, and their time of departure. Second is the tactical decision of route choice from start point to end point via any mid-journey destinations required. Finally, operational decisions are micro-scale behaviours such as avoidance of other pedestrians [35].

The approach to strategic decisions is the appropriate assigning of data and will be discussed later when calibration is covered. Tactical decisions will also be treated in this manner. The operational level must follow a route defined by the tactical level. This route avoids all static obstacles that have been defined by the model environment, but will not consider moving obstacles such as other pedestrians. This can be achieved via the social force model or grid-based models [33, 34].

Grid-based models utilise cellular automata which move around a discretised grid [36]. At each time step these automata will move to an adjacent cell according to a Von-Neumann or Moore neighbourhood, see Figure 2a. The movement is decided through a set of situation-based rules calculating probabilities of moving to each possible position. This method is simple and effective for static environments but is limited in crowd situations [33].

The social force model which models pedestrian movement as the combination of three forces [37], see Figure 2b. The total force for a given pedestrian is calculated and used to generate a trajectory, ω_i , using Equation 1 [37]:

$$m_i \frac{d\omega_i}{dt} = \mathbf{F}_i^0 + \sum_j \mathbf{F}_{ij} + \sum_w \mathbf{F}_{iw} + \epsilon, \quad (1)$$

where $\mathbf{F}_i^0$ is an attractive force in the desired direction; $\mathbf{F}_{ij}$ is a repulsive interaction force between pedestrians i and j ; $\mathbf{F}_{iw}$ is a repulsive force from any environmental features; m_i is the pedestrian's mass; and ϵ is a stochastic error term. At each time step the trajectory of the pedestrian is calculated and moved into a new position on a Euclidean plane.

The social force model was selected for adoption for operational decisions as heavy crowding is expected in station models which would be a limitation for the grid-based method.

2.5 Software Review

An ABM could be developed in any modern coding language, however, there are many dedicated softwares available. These can be split into two categories: toolkits and high-level languages [38].

From an overview of the agent-based modelling softwares available [38, 39], the selection was narrowed to AnyLogic [40] and NetLogo [41]. TfL use LEGION SpaceWorks for station modelling [42], however, this software is subscription based and was not available from the university.

AnyLogic is a subscription based software for which an academic license was available. It is a toolkit type software using an object-oriented Integrated Development Environment. The modelling approach is limited by the libraries available [43], however, it offers much greater scalability for this cost [44].

NetLogo is a high-level programming language approach. It is open source and more flexible in approaches which can be implemented due to not being limited by the libraries available. It is a popular choice for research on ABMs [45]. However, its scope is limited by its high-level programming approach which leads to limited scalability [46].

AnyLogic was chosen due to the mitigated risk of scalability limitation, previous experience with the software meaning less difficult adoption and TfL using a similar software means that the development is more applicable to the company.

3 Network Data Analysis (OP)

ABM has been identified as the modelling technique of choice to simulate passenger interactions and disease spread. However, with 268 independent underground stations to investigate in the TfL network, the use of ABM to model each of them is infeasible within a sensible timescale. By analysing network data it is possible to link stations based on network trends and hence broaden the range of applicability of the results from the ABM.

3.1 The NUMBAT Datasets

The TfL network data for passenger movement used in the course of this project is the NUMBAT 2019 data [47]. The proper meaning of the abbreviation is unavailable in the public domain. NUMBAT data provides the details of passenger flows, entries, exits, boarders, alighters and train frequencies for every link and station in the TfL network in 15 minute intervals across a 24 hour period for either: Monday through to Thursday (averaged), Friday, Saturday or Sunday. This data is designed to represent travel demand on a typical autumn weekday [47]. The simulations and calculations throughout the course of this study utilise the NUMBAT dataset to provide passenger flow details.

3.2 Categorising Stations via Busyness

There are multiple different factors that affect the likelihood of transmission of COVID-19 in both TfL underground stations and train carriages. Examples of these are ventilation, use of masks, station topography and the cleaning protocols employed by TfL. This section of the report will focus solely on

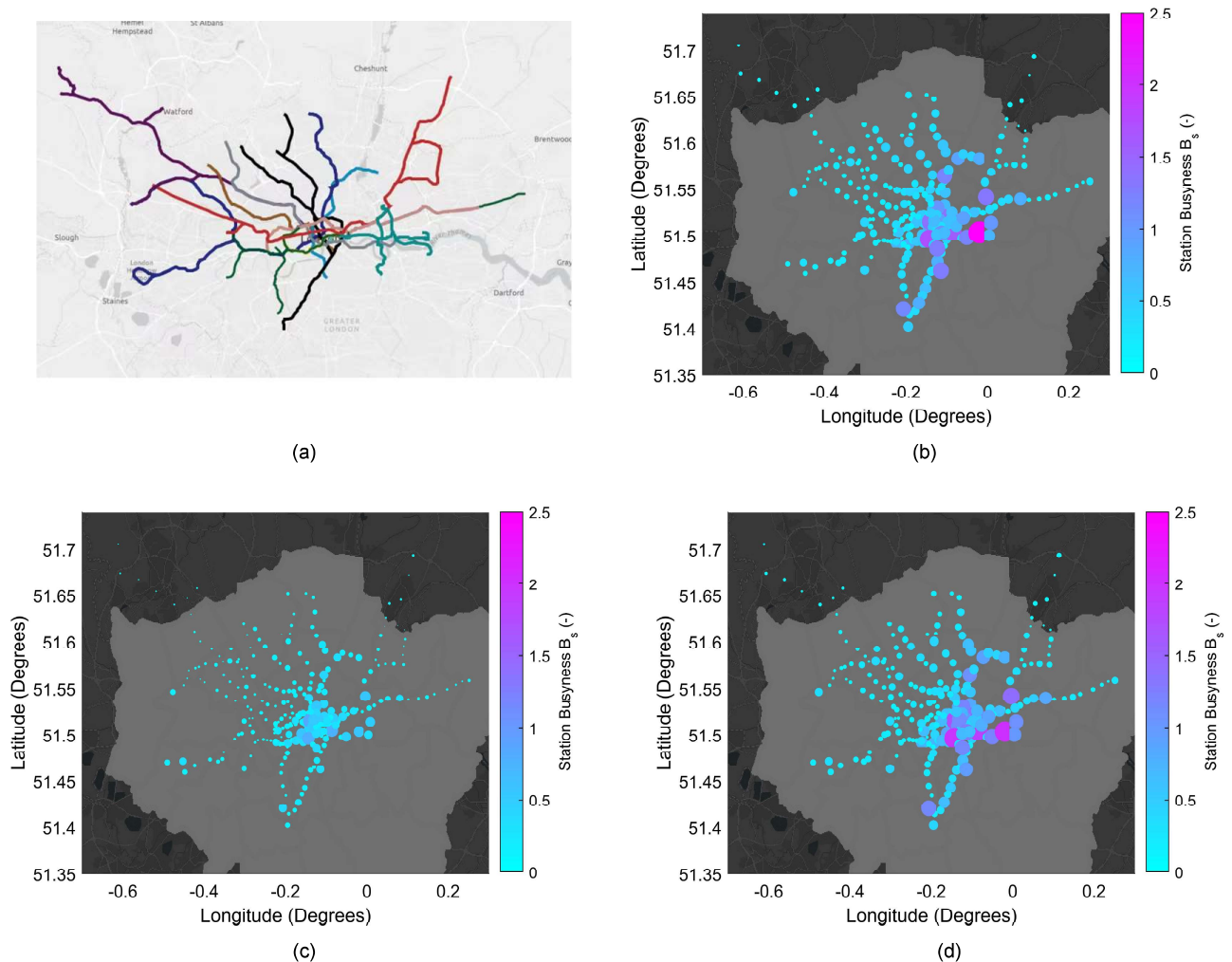


Figure 3: Geographical representations of the busyness at stations around the TfL underground network. Figure 3a shows a geographical representation of the TfL network for reference [48]. In Figures 3b, 3c and 3d the light grey area represents the boroughs of London. Figure 3b shows the busyness during the morning busyness peak (8.30am). Figure 3c shows the busyness during lunch (1pm) and Figure 3d shows the busyness during the afternoon peak (5.30pm).

the busyness characteristics of stations in the network. As the only influence on COVID-19 transmission which is a non fixed entity, busyness presents a means by which to characterise each station and identify passenger flow similarities between stations accordingly. Further to this, by studying the trends in the busyness of stations, predictive models can be established which can be used to develop more effective means of mitigating against infection in a station. Suggested forms of mitigation include the rerouting of passenger flow (explored in Section 9) and the allocation of cleaning schedules to synchronise with the fluctuations in busyness.

Busyness for the stations is defined as the passenger population density. A proxy value must be applied in place of station dimensions as these values are not in the public domain. Busyness is defined by Equation 2:

$$B_s = \frac{N}{P \cdot T}, \quad (2)$$

where B_s is the busyness of each respective station, for the given NUMBAT intervals, N is the number of passengers in the station in each interval, P is the number of platforms at each station, used as an indicator of the size of the station and T is the average capacity of the trains used at the relevant station.

Figure 3 provides a visual for where in the network facilitates the greatest infection risk due to busyness. It is clear that for the peak times the entire network experiences an influx in busyness, the focus of which is directed at central London stations. In order to provide the maximum result for reducing risk of transmission, TfL sanitation strategies should focus on minimising infection risk at these times.

3.3 Busyness Clustering

Station busyness patterns and trends can now be analysed. Station busyness throughout the day is unique to each station. However, due to trends in human behaviour many station busyness profiles share common profile features. Clustering presents a means of grouping these stations by using density patterns exhibited in the NUMBAT dataset [47]. Clustering is widely applied in the data mining and machine learning communities for tasks such as pattern recognition and data processing [49]. In this context it is an effective tool as it enables the ABM results from a select few stations to be applicable across a range of stations which share similar busyness features.

The NUMBAT data provides 96 data points for each of the 268 stations. Clustering for all these points would have been impractical, so to create a more manageable dataset, the data was smoothed to avoid localised maxima and minima and then the data points corresponding to the remaining maxima and minima were extracted.

There are multiple clustering algorithms which are commonly applied in discrete data processing. Of these a few can be seen to be suitable for the application to the task of grouping maxima and minimas. These were: K-means, K-medians, Agglomerative and Divisive clustering.

Agglomerative and Divisive clustering are both forms of hierarchical clustering. Clusters are represented through the use of a dendrogram at varying levels of granularity [49]. Divisive clustering explores a top down approach in which the dataset starts as one large cluster and is then successively partitioned until a tree like structure forms. Agglomerative clustering explores the opposite. In both divisive and agglomerative clustering a value is assigned as either a maximum or minimum distance between clusters to use as a cut off value. Aggarwal et al. highlights a disadvantage in these algorithms in that cluster imbalances and inconsistencies in cluster sizes (such as 'chaining' in agglomerative algorithms) can be difficult to avoid [49].

K-means and K-medians are both forms of flat clustering where the data is divided into multiple clusters immediately. This is often performed using random allocation of centroids or via a partition representative [49]. A distance metric is then calculated for all points in the data set to their nearest allocated centroid. From here either the mean or the median of the collective distances is determined and used to relocate the centroids. Aggarwal et al. highlights the advantages of these two methods lying with their simplicity [49]. The disadvantages lie with the initial allocation of centroid locations. If the dataset is noisy or multiple outliers exist, then both the k-means and k-medians clustering techniques can converge into these groups of outliers.

A decision was made to proceed with K-means as the clustering algorithm. This is because it is simple to implement and exists as a function in MATLAB [50]. In order to avoid the clustering algorithm converging inappropriately on localised groups of data points as previously discussed, the K-means

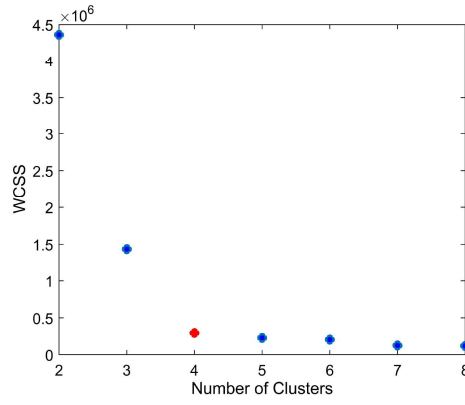


Figure 4: An elbow plot showing the within cluster sum of squares (WCSS) score for a range of clusters for the maxima dataset. The red point corresponds to the number of clusters used for clustering the busyness maxima data.

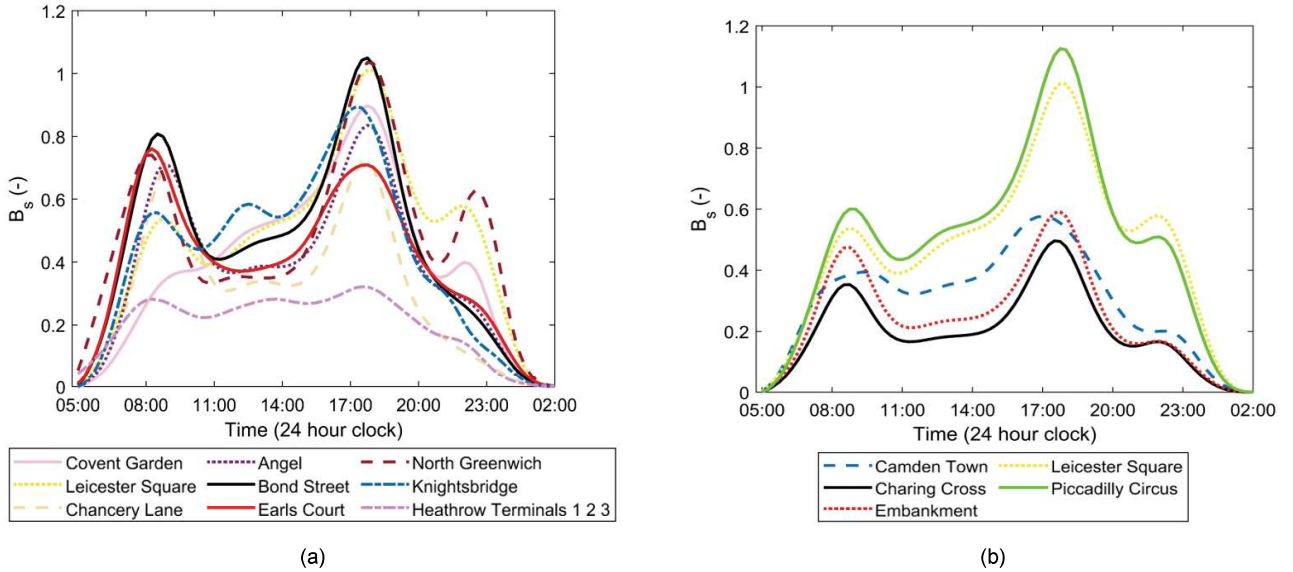


Figure 5: Busyness plots for the Monday to Thursday NUMBAT 2019 data. Figure 5a shows the 9 distinct busyness patterns which represent the possible profile shapes for all the stations in the network. Figure 5b shows an example of a cluster of 5 stations gathered based on their similar minima and maxima locations.

algorithm is repeated multiple times and the most common convergence points are chosen. This is shown Equation 3:

$$J = \sum_{j=1}^k \sum_{i=1}^n ||x_i^{(j)} - c_j||^2 \quad (3)$$

where J is the K-means objective function, k is the number of clusters, n is the number of data points, $x_i^{(j)}$ is the individual data point corresponding to a maxima or minima in a stations busyness profile, i represents the specific maxima or minima, j represents the cluster and c is the relevant cluster centroid.

Implementation of the K-means algorithm requires careful selection of the number of centroids. The elbow method is used for this task. The elbow method uses the within cluster sum of squares (WCSS) distance between a cluster centroid and its member data points across a range of assigned numbers of centroids [51]. This method helps to establish a computationally efficient number of data points to use which will effectively cluster the data points. The number of clusters assessed ranges from 2 to 8.

From inspection of Figure 4 it can be seen that 4 clusters would be the appropriate choice for clustering the maxima values. At this point on it is evident that an increase in the number of clusters yields minimal improvement in the WCSS, yet could prove computationally expensive. The elbow method was subsequently repeated for the minima in the data set and the number of clusters was chosen to be 5.

By comparing the stations to see which shared maxima and minima within the same clusters, groups of stations sharing similar busyness profiles could be established revealing 9 main types of significantly different busyness profiles. The size of these groups of stations varies dramatically ranging from 155 stations to 1. The profiles from a station in each of these groups is illustrated in Figure 5a.

4 Approach To Model Development (JH & WA)

All models are an abstraction of the system they aim to simulate. Only the dynamics needed to produce a model that is valid for obtaining the results desired require modelling [5].

The modelling approach gives a starting point for the level of abstraction. ABM can be used on a wide array abstraction levels, from mid-high to the lowest level, micro-scale simulations [5]. As policy that affects the behaviour of passengers down to operational level decisions, a low level of abstraction is required. However, the data available to an outsider to TfL is limited for security reasons. As a result, its usage is limited in this situation, instead providing a proof of concept for the process.

The target is to create a model of how disease is transmitted through a LU station at the micro-scale level. This model will need to provide insight into the ‘risk’ that the station poses to a passenger and how this changes under mitigation policy. Risk will be defined as the probability of a susceptible passenger entering the station becoming infected as a result. The approach to this will have the aim of being non-specific, instead providing a framework.

The model will be split entirely into two sections which will integrate to complete the model. These are the disease transmission model and station model covered in Sections 5 and 6 respectively. The integration process will be covered in Section 7. This split acts as an additional proof of concept that a disease transmission model can be inserted into an already existing agent-based station model.

5 Disease Transmission Modelling (WA)

As an agent based approach has been adopted, the probability of infection for each agent, dependent on the agents actions within the model, is required. There is focus on a parameterised model that can be used for any disease and varying situations.

5.1 Dose Response Model

To model the probability of infection posed to an individual agent, a dose response model is utilised. The transmission probability for a particular agent is determined by the stochastic exponential dose response outlined by Equation 4[25] :

$$P = 1 - e^{\frac{-d}{\text{HID}_{50}}} , \quad (4)$$

where d is the total dose received over the agent’s time in the model and HID_{50} is the median dose to required to cause infection in 50% of the population. The exponential form presented is chosen as it characterises the dose response relationship for COVID-19 well [52]. To produce a broadly applicable disease transmission model, other dose response relationships, such as the exact Beta Poisson, have been implemented. This allows for other transmissible diseases such as Filovirus (Ebola) and Influenza to be more accurately modelled.

The dose response model also allows for flexibility with regards to units of viral quantification. This is important as HID requires a long time to quantify due to disease lethality and ethical considerations. In early pandemic scenarios other metrics can be used, based on the available data. It also allows for a comparison of dose received. This is significant as the level of dose received often has an impact on disease severity, incubation time and viral load [53]. This suggests potential for disease severity mitigation as well as the mitigation of further infections.

5.2 Agent-Based Approach with the Dose Response Model

The disease transmission model has been implemented for use within agent-based modelling approaches. Three key transmission routes are considered: short-range aerosol and droplet, long-range aerosol and fomite transmission. Their individual dose contributions are summated over an infected agent’s time in the model. The dose response model is then used to calculate their infection probability using Equation 4. A compartmental structure is used to manage the infective condition of each agent.

5.2.1 Agent State

The model follows a SEIR compartmental structure, categorising each agent in to a state of susceptible, exposed, infected or recovered. Upon entering the model an agent is randomly assigned a value between 0 and 1, this value determines the state assigned dependent on intervals determined by a parameterised population distribution allowing for the model to reflect pandemics at different phases. The statechart in Figure 6 shows all the possible states the agent can have within the simulation and the transitions between these states.

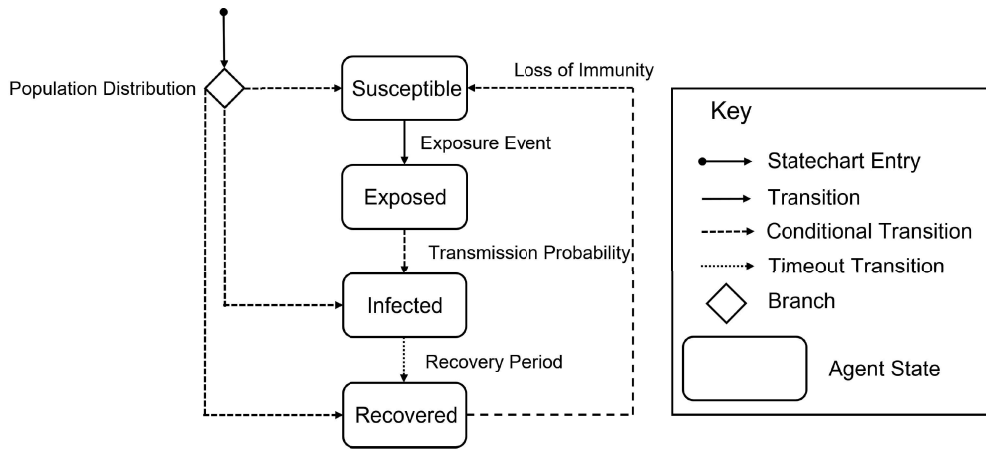


Figure 6: Agent statechart diagram

The transition between the susceptible and exposed agent states is dependent on a dose receiving interaction through any of the transmission routes. Transition between the exposed state and infected state is dependent on an incubation period and infection probability. The infection probability is found using the dose response relation in Equation 4. Due to the model run time being less than a day, and most disease incubation times being substantially longer than this, upon leaving the model the infection probability is used to determine whether infection occurs. For a similar reason, the recovery period transition and loss of immunity transition are also outside of the model run time.

As ABM is used, each infective agent is assigned varying dose parameters viral load concentration C_v and hand dose H_d . These are representative of the concentration of RNA copies present in their saliva and present on their hand respectively. This accounts for the range of values present in a population due to personal behaviours and stage of infection [25]. To assign these values, parameterised distributions are used.

5.2.2 Short-Range Aerosol and Droplet

Short-range aerosol and droplet transmission dominate interactions where interpersonal distance is less than 2m [25]. The dosage received is a complex situation dependent on the interaction time, distance and several agent characteristics. To integrate this in to the ABM, the model utilises AnyLogic's in-built distance based network and messaging system. This allows for agents within a set distance to be added to a network. As AnyLogic's network system does not automatically refresh, an event is used to periodically check which agents are within 2m of an infected agent.

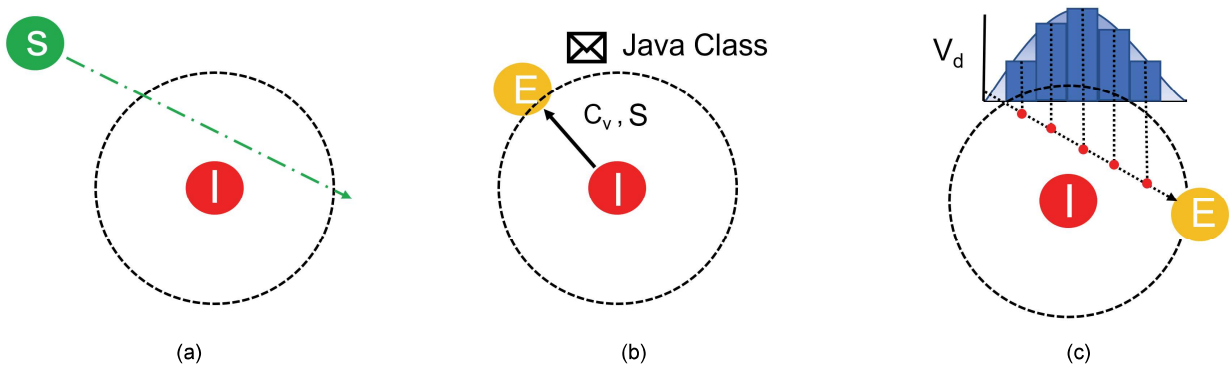


Figure 7: The short-range transmission route mechanism during different stages of an interaction, Figure 7a before interaction, Figure 7b shows the Java Class messaging system used, Figure 7c shows the mechanism regarding the calculation of dose via discretised integration throughout an interaction.

Within the network, Java Class messages are sent from infected agents relaying their C_v and distance. This message is sent for every event cycle within the set range. The distance from the infected agent greatly affects the potential received dose, several papers investigate the relationship between distance and dose utilising CFD. These results have been consulted to assess the potential droplet volume distribution and total droplet volume inhaled per unit time, V_d , by an agent at a particular interpersonal distance. The results of Cortellessa [25] have been taken and line fitted to construct a distance

dependent equation for a likely V_d . The received volume of droplets varies exponentially in agreement with literature [54]. The received droplet volume with respect to distance and the total received dose are calculated via Equation 5 and Equation 6 [25] respectively:

$$V_d = 4.67e^{-8.94S} \quad (5) \quad d_{sr} = (1 - \eta_m)V_d C_v \Delta t, \quad (6)$$

where S is the distance from the infector, d_{sr} is the dose received, η_m is the filtration efficiency of a mask, V_d is total droplet volume inhaled per cycle time, C_v is viral load and Δt is the event cycle time. Mask wearing is implemented into the model by taking the droplet filtration efficiency of the mask of a particular agent, if an agent is not wearing a mask, this is set to 0. The total short-range dose is then equal to the summation of the dose received during these interactions and is calculated at the agents removal from the model.

5.2.3 Long-Range Aerosol

Long-Range transmission is primarily concerned with the steady accumulation of small droplet aerosols of sizes less than $10 \mu m$ in enclosed spaces. The dose added to the environment is dependent upon the number of infected agents within. At each timestep an infected agent releases a dose. Similarly, for each timestep each susceptible agent within receives a dose. These dosages are calculated using Equations 7 and 8 [55] respectively,

$$d_{in} = C_v V_e E_r \Delta t \quad (7) \quad d_{out} = C_a I_r \Delta t, \quad (8)$$

where C_v is the viral load of the infected agent, V_e is the volume of aerosols per litre of exhaled breath, E_r is the exhalation rate of the infected agent, C_a is the concentration of pathogen in the environment's air, I_r is the inhalation rate of the susceptible agent and Δt is the model time step. A susceptible agent's dose received is summated over their time in the environment.

Removal of dose from the environment is dependent upon: the ventilation rate of said environment in air changes per hour; and the inactivation of the pathogen in air dependent on a half life relationship. This removal is calculated every time step. The total dose within the environment is assumed to be well mixed and results in a uniform concentration of virus in the environment.

5.2.4 Fomite

Fomite transmission occurs when pathogens are transferred via a shared surface. These events occur primarily at services within the station. The dose transferred to the surface and dose transferred to a susceptible agent are described by Equations 9 and 10 respectively [28],

$$d_{tf} = \eta_{ts} d_h \frac{A_s}{A_h} \quad (9) \quad d_{rf} = \eta_{th} d_s \frac{A_h}{A_s} \eta_{HTF}, \quad (10)$$

where d_{tf} is the dose transferred to the surface in an interaction, η_{ts} is the transfer efficiency of contaminant from hand to surface, A_h is the area of contact, A_s is the surface area of the fomite, d_{rf} is the dose received by an agent, η_{th} is the transfer efficiency of contaminant from surface to hand, d_s is the dose present on the fomite surface and η_{HTF} is the transfer efficiency of hand to facial mucosal membrane. In each interaction logic was added so that the ratios of areas are limited to one representing the maximum dose transfer that can occur.

The dose on the surface of the fomite agent and on an agent's hand dissipates at a rate dependent on the half life of the contaminant. This is again applied per model time step. The contaminant can also be removed from the fomite agent or agents hand through cleaning events.

5.3 Model Validation and Verification

Model verification ensures the model operates as expected so outputs hold confidence. The disease transmission model has been verified through the use of several micro models, each verifying a key transmission route. Input parameters within each of the models are changed and the trends in the outputs, are verified against expected system behaviours. Whilst several of the inputs for each of the transmission routes have been varied, for the sake of brevity, a selection has been displayed to prove the key aspects of each transmission mechanism. The mean probability of infection is defined by the number of new infected agents divided by the number of susceptible agents that have passed through the simulation. The outputs have been run with multiple iterations on random seeds to reduce the impact of stochasticity on the trends.

5.3.1 Short Range Aerosol and Droplet Verification

Within the micro model, the only transmission route enabled is the short-range aerosol and droplet transmission. Agents move in random directions interacting within the environment.

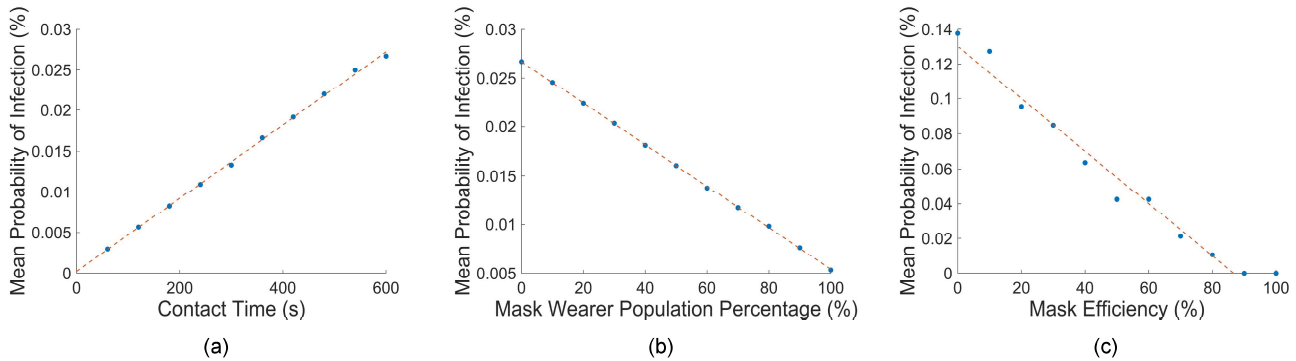


Figure 8: Verification plots for the short range and aerosol transmission route. Figure 8a, mean probability of infection against average contact time, Figure 8b mean probability of infection against the percentage of the population wearing masks and Figure 8c the mean probability of infection against the droplet filtration efficiency of the masks.

Figure 8a shows a positive correlation between average contact time of the agents and the mean probability of infection. This is to be expected as the dose received is dependent on the amount of time the agent spends within range. The linear trend is a result of the dosages received being low enough such that the exponential dose response curve is within a linear section. Figure 8b shows the negative correlation between the percentage of the population wearing masks. Again the linear trend is expected as the low dosages result in a linear relationship between dose and probability of infection. Figure 8c shows the negative correlation in the mask efficiency of the population, this is expected as the total dose prevented from reaching a susceptible agent scales with the mask efficiency.

5.3.2 Long Range Aerosol Verification

The micro model used to verify the long-range aerosol transmission route consists of a number of agents occupying a shared volume of space. The time spent within the environment, number of agents within the environment, inhalation and exhalation rates of the agents were altered. The total model environment dose and dose concentration is displayed to perform visual verification and ensure the removal of virus, via deactivation and ventilation, is performed as expected.

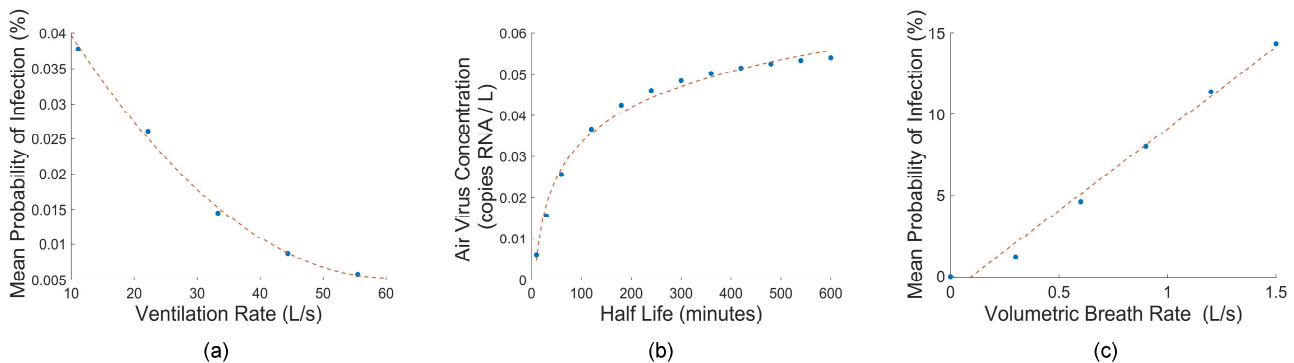


Figure 9: Verification plots for the long-range aerosol transmission route. Figure 9a, mean probability of infection against ventilation rate, Figure 9b mean probability of infection against the percentage of the population wearing masks and Figure 9c the mean probability of infection against the droplet filtration efficiency of the masks.

As can be seen in Figure 9a, altering the ventilation rate within the model has a clear effect on the mean probability of infection caused by the long-range transmission route; this is to be expected as the concentration of the aerosolised virus within the space cannot reach significant levels in situations with higher removal rates. Figure 9b shows the impact of varying the half life of the suspended particles. The result shown is to be expected, as the half life of the contaminant increases, the removal rate decreases and for the same model time the concentration of viral copies approaches a constant level. Figure 9c shows the trend expected when increasing the volumetric exhalation rate of the infected agents. This is due to the increased viral production within the space.

5.3.3 Fomite Verification

A micro model with five ticket gates acting as fomite agents was created to verify this mechanism. The dose present on each agent's hand is displayed alongside the dose on each gate to provide visual verification of the process, additionally numerical checks were performed to ensure the total dose input to the system was equivalent to the total removed, either through transfer or dissipation according to the half life of the contaminant. In a similar procedure to previous verification, the parameters involved with this particular transmission route were varied to ensure the model functioned as expected.

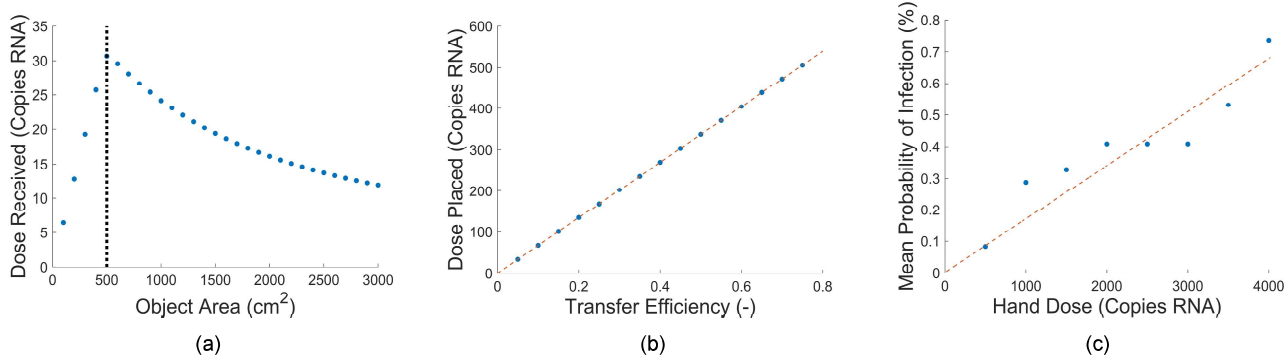


Figure 10: Verification plots for the fomite transmission route. Figure 10a shows the dose received by an agent against object area, Figure 10b shows the dose placed against transfer efficiency and Figure 10c the mean probability of infection against agent hand dose.

Figure 10 shows the plots used to verify the fomite transmission mechanism. Figure 10a shows the relationship between the object area and the dose received, the dose received scales linearly with the object area before reaching the size of the agents hand at 500cm^2 . Up to the hand size of the agent the available dose scales with the object size as the area for possible transfer scales. Beyond this point the dose becomes distributed over the larger area, and becomes less concentrated and therefore when a contact event occurs an agent receives less dose. This follows one over recursion as predicted by Equation 10 when hand area is kept constant. Figure 10c shows a roughly linear trend. The variation in this trend is assumed an effect of the stochastic present in the model. As is to be expected the mean probability scales positively with increased dose on an infected agents hand. The transfer efficiency (shown in Figure 10b) also follows a similar trend with is to be expected and is forecast by the relationship in Equation 9.

5.3.4 Validation

Model validation ensures the values outputted by the model are accurate to the degree required. Due to the situational nature of the probability of infection, finding reliable comparable data to validate the model against is difficult. Therefore, validation is approached via the analysis of super spreader events. Super spreader events are situations where a disease's situational probability of infection is reported far higher than normal, often attributed to the viral load and behaviours of the super spreader [56]. For this reason, the data for these events is more readily available.

To validate the disease transmission model, micro models are created with input parameters mimicking super spreader events, Monte Carlo analysis performed with 100 replications and the range of possible mean probability of infection compared to the case study. The input for C_v is chosen to be the 99th percentile value of recorded copies per ml of sputum samples as super spreader events suggest infectors with much higher viral loads[25].

Three outbreak events are considered and used to validate the model, an outbreak in Ningbo, China, and two buses in Hunan China from the beginning of the COVID-19 pandemic. Information regarding relative volumes and ventilation rates is provided [57]. Unfortunately, due to the little known information regarding touch behaviour, the validation of the fomite transmission route is not possible. Therefore, only the short-range transmission and long-range aerosol routes are considered.

The results of the validation study can be seen in Figure 11. The outbreak data does not show complete agreement with the model outputs with the model predicting anywhere between a 66% difference and 12% difference in the median probability of infection. The Hunan Bus 1 case study data falls comfortably in the range of possible mean infection probabilities output by the situational micro model. Compared to each case the median model output is lower than the actual probability of infection, this is

suspected to be caused by the lack of fomite transmission mechanism present within the model. The varying difference can be explained by the variance in fomite causing behaviour by the individual which is hard to quantify.

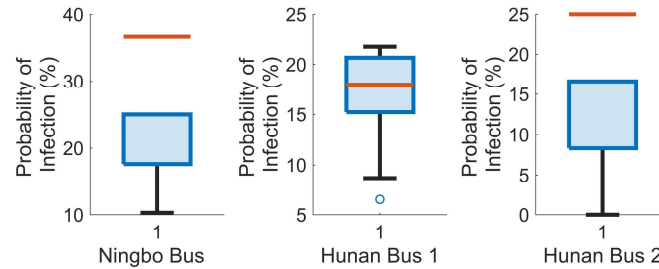


Figure 11: Model output probability of infection for three case studies (blue): The Ningbo Bus, Hunan Bus 1 and Hunan Bus 2 along with the actual case study data (orange).

To provide additional validation, the model is also validated through use of a proxy measure, the R number. A diseases R number is defined as the average number of secondary infections caused by an initial infectious case [58]. Although this method does not validate the probability of infection output by the model, the R number, a number highly correlated to the probability of infection can be used to validate the model.

A station model of Vauxhall was run over the course of the day returning a median R value of 0.53. Over the infectious period of COVID-19 (4 days), assuming two station situations a day (typical for commuters) results in an R number of 4.24 which falls well between the range of R numbers reported by literature 1.90 to 6.49 [58]. Although this might suggest the model outputs a higher R number than reality, when an individuals interactions beyond the station are considered, public transport is reported to be a major vector for transmission [1] and it is not unreasonable to assume the vast majority of an individuals interactions are experienced on public transport.

Although providing a level of validation, the confidence provided by each of these methods is not complete. The large difference in presented super spreader values and model outputs suggest that the models risk values contain reasonable uncertainty. Despite this, the tools use in a comparative manner means the absolute values of risk are less important and the risk difference between stations and policy, when implemented, can still be used to make informed decisions to reduce risk in the most applicable situations.

6 Modelling Movement within a Station (JH)

Attention now moves to the modelling of the station itself. This will take the form of an hybrid agent-based and discrete-event model using the methods discussed in Section 2.4. Station parts will be modelled, calibrated and validated.

6.1 Modelling Station Parts

An urban rail station is an environment designed for the efficient movement of passengers from the street onto trains and vice versa. As such they typically contain features limited those required for this movement. Some stations may contain other components but these have been classified out of the scope for this project due to lack of data.

Figure 12 provides an abstract station layout and the possible routes through it when moving from the street to the platform. There are 144 possible routes using a total of 19 different station features. The modelling approach taken must be able to easily and robustly model these complex problems.

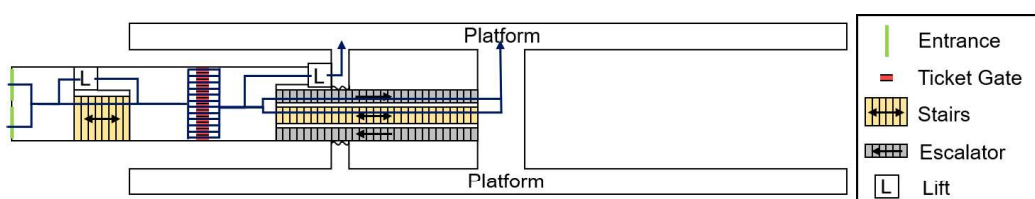


Figure 12: An abstract diagram of Brixton LU station with all possible routes through the station to a platform.

To solve this, a framework approach was taken whereby all types of station facilities were formulated into blocks dedicated for modelling them. The selection of parts required were decided as: entrance/exits, ticket barrier lines, stairs, escalators, lifts and platforms. With each of these modelled in part, any station should be build-able from their correct combination. Figure 13 provides an example using the station in Figure 12. As AnyLogic was selected as the software for model creation, the built in pedestrian library was used. This has inbuilt social-force model capability and contains generalised functions that aid in model construction, however, some facilities required a more abstract use of these functions to achieve the desired dynamics.

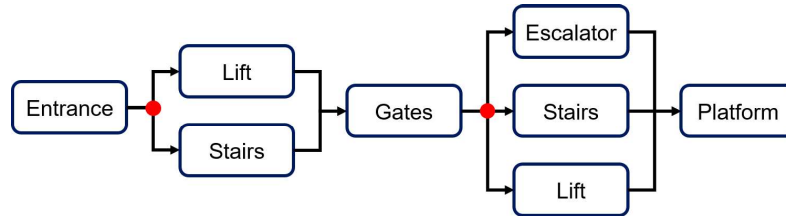


Figure 13: How the movement described in Figure 12 can be modelled using the framework approach. Red dots highlight the decision points discussed in Section 6.2.

6.1.1 Entrance/Exits

Stations typically have multiple ingress and egress points. As these are one of the two points that pedestrians enter or exit the model, there must be an ‘attractor’, a point that the agents reach before being removed. There are two ways to approach this. The agents can aim for an area outside of the station with the door modelled as a gap in a wall, or, doors can be modelled as set length arrival lines that act as agent attractors.

The benefit of the latter is that the scope of analysis ends at exactly the threshold for the station, however, interactions just outside the station could be desired in some cases. It is recommended to use arrival lines unless these interactions are specified to be important by stakeholders. This method was used for all models in this report.

In Section 6.2 data for how many pedestrians arrive in a given time will be extracted from network usage data. From this an arrival rate of pedestrians per unit time can be calculated. However, pedestrians do not arrive at destinations perfectly inter-spaced in time. Pedestrian arrivals are commonly modelled as a Poisson stream of events [59]. This was achieved via unique events with an exponentially distributed inter-arrival time. The mean of this was set to the inverse of the rate.

6.1.2 Ticket Gate Lines

Gate lines are an example of a ‘service’. The main stage is a delay period that represents using the service, in this example operating the proof-of-payment device. This will usually be preceded by a form of queuing system to act as a buffer. This behaviour, defined by queuing, waiting and servicing, is an example of where DES crosses into this agent-based model. While the actions of individual entities will be modelled using agents, many station features utilise DES.

A framework for services is available in the AnyLogic pedestrian toolbox. This simplifies implementation as the dynamics of a ticket line can be simulated using a linear service with bi-directional, linear queues. Applicable delay times will be discussed in Section 6.2.

6.1.3 Stairs, Escalators and Lifts

There are three methods utilised to change level at an Underground station. The first is stairs. Stairs are a feature of the physical environment with no dynamics, therefore, modelling of them remains within the agent-based domain. Stair areas can be defined as a sloped environment feature in which, on passing through, agents change level.

Escalators have more complex dynamics than stairs. As such, a method of modelling these features is available in the pedestrian modelling toolbox. This requires specification of parameters defining the dimensions of this escalator and the speed at which it travels.

A different approach is taken for lifts. The structure of lifts is similar to the service defined earlier and so returns to DES. They have many more states and decision points than the gate services considered earlier so the block layout of this feature is complex. To simplify integration, its components were assembled into its own static agent that could be inserted as a custom block into the model.

6.1.4 Platforms

When considering platforms, the dynamics of train boarding was also considered. They are defined by a large waiting area for the majority of time. Upon the arrival of a train, each door of the train results in a repeated set of activities: passengers disembarking followed by boarding.

The platform could be modelled as one area in which passengers wait, from where, passengers move to the closest door upon the arrival of a train. Alternatively, the platform can be pre-split into waiting areas that are closest to a certain door. This approach was chosen as it allows for passengers to be distributed along the platform in a specified way, if it was observed that passengers preferred to wait in certain locations along the platform. As this data was not available passengers were uniformly distributed across these areas.

Upon the arrival of a train a series of discrete events occur. Firstly, the passengers that are disembarking from the train are distributed across the set of train doors in the same processes outlined for the wait areas. At the start of the disembarking-boarding procedure these passengers are inserted into a service queue which models the queuing and using of the door. Once serviced, these passengers are generated on an arrival line representing the door. These lines are positioned and sized according to the dimensions of trains that arrive at that platform. This servicing continues until all passengers waiting to exit that particular door have disembarked.

Once passengers have finished disembarking, boarding at that door then begins. Waiting passengers are told to go to the arrival line and are subsequently removed from the model. However, there are multiple caveats to where and when this process occurs. Passengers are assigned a door when they enter the platform area. This is their desired door but they may change if a train arrives before they are able to reach that point. Additionally, passengers that arrive at the platform when a train is present will also board via the closest door. Therefore, if an agent is within the process of moving to a train door and a boarding event is started, they will move to the closest door and board. If they do not reach the door before the train departs they will continue moving to their original target location.

6.1.5 Station Environment

It is important to define where the limit of the environment in which agents must stay to ensure validity is not broken by impossible routes through the system. Any physical object that acts as a limitation to a pedestrian's walking needs to be modelled to ensure a robust abstraction of the environment. The most common is walls but there could be additional pillars, barriers or vendor stands [42]. Many of these will be included in engineering drawings of the station which can be imported into the model as a template. As these drawings are not publicly accessible for security reasons, station layouts and dimensions were assumed from online photos and simplistic axonometric representations [60]. While this process is not robust, its use is still valid for this proof of concept.

6.2 Calibration

Model calibration is the process of defining parameter values so that the model closely follows the dynamics of the real system [33]. These have been separated into those that define passenger flows and those that define passenger behaviour within the station.

6.2.1 Passenger Flows

Passengers can enter or exit a station through either entry ways or on a train. Data on these passenger flows are available in the NUMBAT data but its form must be augmented for importation into the model. Data that defines a pedestrians starting location and target point is part of the strategic decision level. This is often stored in OD matrices which list starting locations along the first dimension and destinations along the second, each value in the matrix is then the amount of people travelling between these points. See an example in Figure 14a.

OD matrices are required to describe passenger flows within the station but with some differences from this base. Firstly, passenger flows change over the course of a day (see Section 3). Secondly, flows are not continuous as trains result in pulse arrivals. NUMBAT data could be augmented into a three-dimensional OD matrix with the first two dimensions as previously described and the third as time, see Figure 14b. However, this is not importable into AnyLogic. To solve this the matrix will be split along the origin dimension to form a set of two-dimensional time-destination (TD) matrices, Figure 14c. This is the most logical option as it is often tied as the shortest dimension and also results in a data that is more compatible with AnyLogic.

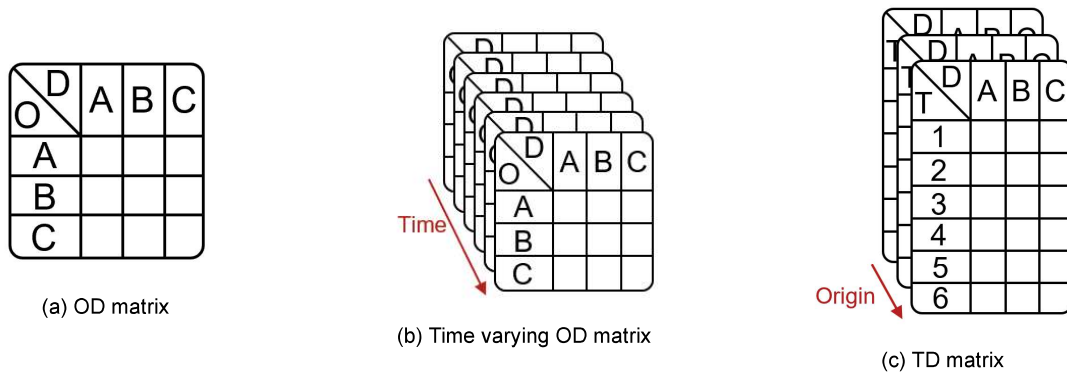


Figure 14: Examples of an OD matrix and the two solutions for including a third dimension to this.

For station entrances the passenger arrivals are kept in the fifteen minute intervals that the NUMBAT data provides. For platforms, arrivals are instead broken into arrivals per train. This is calculated from splitting the number of passenger arrivals in each block over the number of trains that arrive in that interval. The exact timetable is not known so the arrival times were spread equidistantly across the block with one train always arriving at the very end of the interval (provided that at least one train arrives in that interval). To speed up the generation of the TD matrices the process was automated using a MATLAB script.

6.2.2 Station Parameters

It is assumed that layout has been properly imported and the environment defined as covered in Section 6.1. The rest of the parameters depend on the station and the specification of the facilities. For the models created in this project, these values must be assumed, as with station layout. Fortunately, many default parameters for initial model creation are documented by TfL in [42]. This does not include walking or stair ascension speeds for which values used are tabulated in the appendix.

However, parameters for pedestrian route choice within the station are not available. Many stations have multiple routes that can be taken to each point (see the decision points in Figure 13); the decision of these routes would be based on many social factors. If relevant, a study would be considered for calibrating this decision through measuring pedestrian flows. For this project these values must be assumed.

6.3 Station Model Verification Validation

The process of model validation aims to evaluate how closely the model follows reality and how reliable its results are. It is possible for a correctly calibrated model to be invalid and so the process should incorporate data not used in the calibration process [33]. Validation should present multiple forms of evidence. Many rail companies that produce models for pedestrian analysis produce guidance documents that include a validation process [42, 61, 62], although, lack of data again limited the processes that could be utilised. Additionally, verification ensures that calibration has been correctly performed before beginning the validation process.

6.3.1 Visual Verification

The simplest form of verification is to visually study the model in progress, a process made easier through 2D and 3D visualisation. The operator studies the model to see if all aspects are acting as expected [33]. This is a subjective process so numerical methods should be used following this. Visual verification was performed throughout the model creation process.

Another simple check is to ensure that conservation laws have been followed. In this case it is that the number of agents in the model (station) is equal to the number that have entered minus those that have left. This was implemented as a check at the end of the simulation.

6.3.2 Comparison of OD Matrices

An important verification technique is checking that OD matrices were correctly implemented [62].

A system generated OD matrices from the model passenger flows and saved this at the end of the simulation. The average of these was taken for comparison to the inputted values. To compare these values the GEH statistic is calculated [33, 61]. The GEH statistic considers relative and absolute errors to create a comparison that is more independent of the scale of the flow. This is used as the passenger

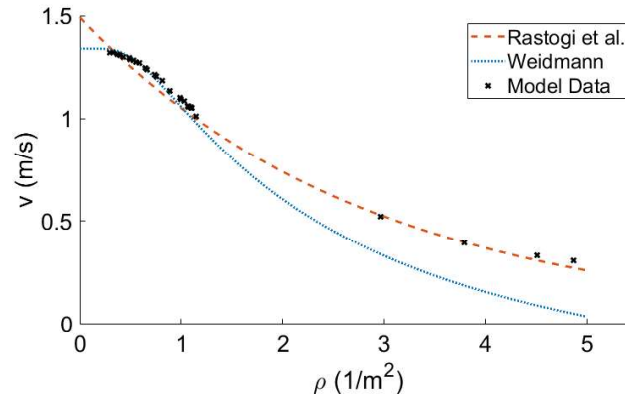


Figure 15: Relationship of average pedestrian speed, v , against crowd density, ρ , for the corridor model and guideline curves presented by Rastogi et al. [63] and Weidmann [64].

flows range from the order of one to hundreds. If percentage error was used, smaller flows may be highlighted as erroneous compared to larger flows due to the same level of variation.

The GEH statistic is calculated using Equation 11 [61]:

$$GEH = \sqrt{\frac{2(M - O)^2}{M + O}}, \quad (11)$$

where, M is the modelled value and O is the observed value. This is calculated for each OD value. From this values below five are considered to be strongly correlating, thus required, with values below three being preferred [61].

The test was performed on a model of Marble Arch station. Taking an average of the collected ODs over ten runs and calculating the GEH yielded in 99.5% of flows to be below three and 100% below five. Therefore, the passenger flows in this station model are deemed to be valid.

6.3.3 Flow Rate/Level Of Service Comparison

A form of data often collected before modelling a system is the behaviour of crowds in key areas. Of particular interest are areas of potential high congestion such as bottlenecks. This may be measured in flow rate, pedestrian speed or level of service [33].

A simple corridor model with a one-directional pedestrian flow was created for which the number of pedestrians passing along the corridor was varied. The average walking speed and pedestrian density in the corridor was measured. This is plotted against two guideline behaviours in Figure 15.

It can be seen that the modelled results follow Weidmann's curve for the initial flows. However, the flow becomes suddenly unstable when passing densities of one pedestrian per m^2 . While the values remain acceptable, this could suggest instability in high density flows.

6.3.4 Sensitivity Analysis

A robust model will be valid for reasonable changes in input. For sensitivity analysis the flow results can be recollected but with a change that should alter this metric. These results can then be compared to ensure that the change is as expected [33]. This process can go as far as shocking the system. Suddenly deactivating half of the gates in a ticket line should suddenly decrease service throughput but it will not break a robust model. A variety of shock tests were performed on the model features outlined in Section 6.1. The model was found to be robust in most instances other than those where routing was impossible, ie, the desired destination lay outside of the defined environment. Although, situations where this occurs are in themselves invalid.

6.3.5 Identifying Limits of Validity

As this report focuses on the LU, two stations at the opposite ends of the scale were selected in order to test the limits of validity. The least busy sub-surface underground station found via the analysis outlined in Section 3 is Wanstead with an average busyness of 0.008. At the other end of the scale, Canary Wharf is the busiest with an average of 0.901. Models for each of these stations were constructed.

The model for Wanstead required minimal calibration to produce expected flows. Canary Wharf posed some difficulty due to the footfall at peak times (this reaches a maximum of 6187 expected passengers

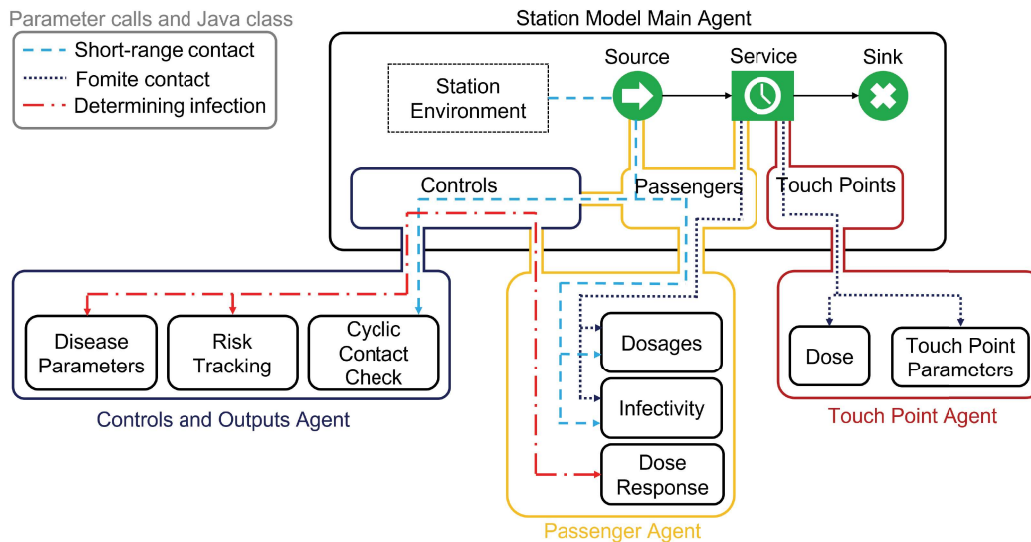


Figure 16: How disease transmission modelling is integrated into a station model using agents.

passing through the station in a fifteen minute period). As such calibration was difficult, compounded by the large number of infrastructure features in the station. Regardless, it is concluded that the approach is valid across the network.

7 Agent-Based Model Integration (JH & WA)

Before disease transmission analysis can be performed on a station, the methods of the previous two sections must be integrated together. As the transmission model was designed so that it could theoretically be inserted into a pre-existing station model, it will be assumed that a station model already exists at this stage. Integration is a two step process of building the model and applying applicable parameters.

7.1 Building a Integrated Model

To facilitate ease of this process, all transmission dynamics are handled within agents. Three agents are provided, a passenger agent, a touch point agent and a static agent that contains many of the controls and outputs. Additionally, a Java class is required for dose data transferal between agents. Figure 16 summarises how agents integrate in a diagrammatic manor.

The passenger agent handles the infection state of an individual passenger as described in Section 5.2.1. The touch point agent is more simplistic as it must simply track is surface dose and hold information on its surface area and transfer efficiency. It is generalised in its form so that it can be applied to any touch point. Finally, as aerosol transmission is only considered within lifts at this stage, the dynamics of this are added to the lift agent discussed in Section 6.1.

To integrate disease transmission into a station model three steps must be performed. Firstly, the agents must be inputted as the agents used for their respective purposes, ie. all passengers inserted will be assigned to be the disease passenger agent and each service will utilise the touch point agent as the resource required for operation (provided that the service in question has a touch point).

Secondly, contact events must be coded. Touch events are simple as these occur within service blocks so can be inserted as an additional function to run when the service is used. Short-range contacts are distance based as and are handled by distance based networks as described in Section 5.2.2. This network approach is applied cyclically and so an event that performs this check is included in the control agent. Finally, long range transmission remains within the lift agent only where dosage calculations are performed at the end of each lift movement. Dosage information is transferred using a Java class; this contains both the total dose transferred in copies RNA and the transmission type so that individual contributions can be compared.

Finally, infection tracking parameters and data sets need to be inserted to aid with analysis. These are contained within the control agent which can be inserted into the model for faster integration.

7.2 Key Model Parameters

For several disease parameters literature reports large ranges in the accepted values. To investigate the the variance in reported values, sensitivity analysis is performed. Sensitivity analysis provides opportunity to ascertain the models reaction to changes in input and ensure the model provides reasonable outputs for these changes. This serves as an indicator to which parameters have the largest impact on the output thus require more extensive research.

As the relative contribution of each transmission mechanism is dependent on the situation of each agent, the results of the sensitivity analysis may change depending on the scenario considered. For this reason, sensitivity analysis is performed on a series of micro models to establish the parameters with the most impact to each transmission model.

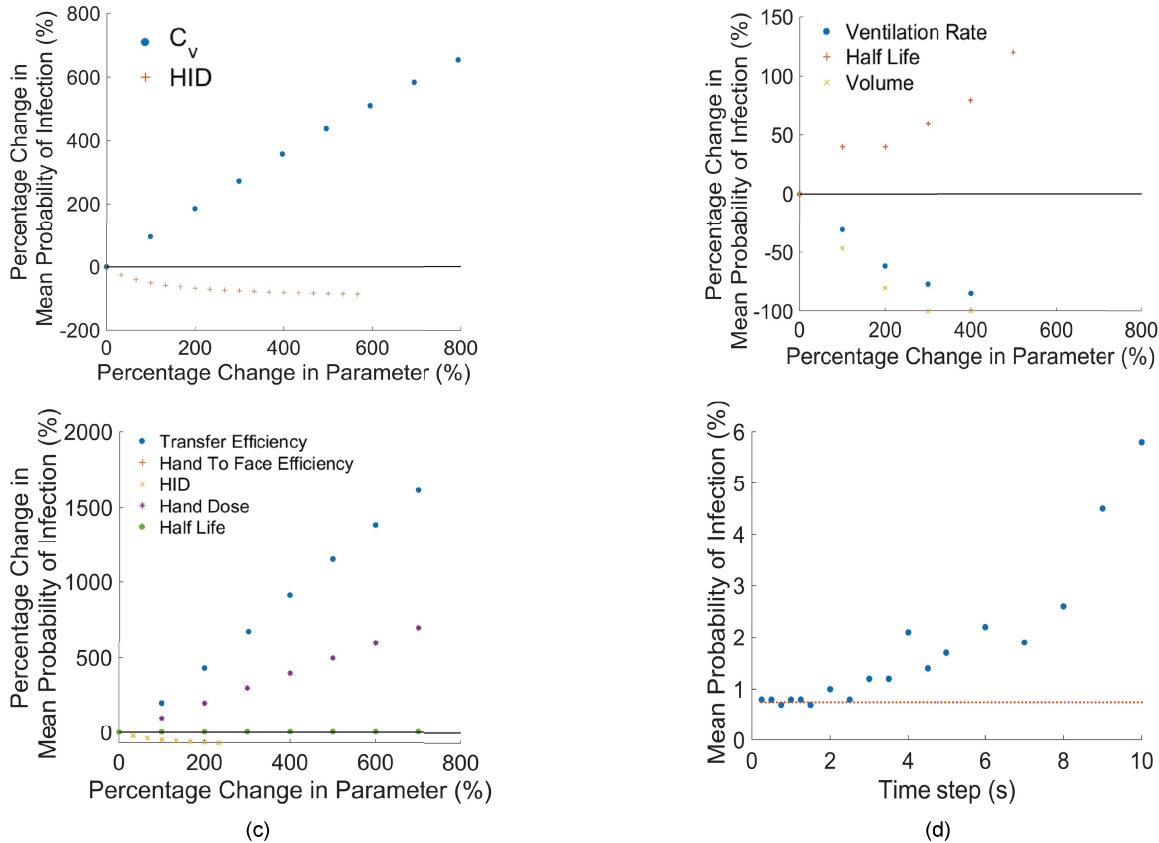


Figure 17: Sensitivity Analysis for Figure 17a short-range transmission, Figure 17b long-range transmission, Figure 17c fomite transmission and Figure 17d timestep analysis

The results are provided in Figure 17. For the fomite route (Figure 17c) these are the dosage found on the agents hands and hand to face transfer efficiencies. For short-range (Figure 17a) this is the viral load of infected agents while for long-range (Figure 17b) all values have a similar, lesser contribution. Therefore the distributions assigned to each of these values must be carefully researched.

A micro model was constructed to assess the effect of varying the model timestep. The model timestep is integral in several disease transmission mechanisms but most importantly for short-range interactions. For this reason the micro model mimics the shortest possible interaction. The timestep was varied and Figure 17d shows the result. It is clear that the mean probability of infection converges as the timestep is reduced below 1 second and provides no additional benefit for the incurred computational cost. This therefore is identified as the limit of resolution.

8 Empirical Station Risk Characterisation (IJ)

From trialling constructions of the passenger flow and disease transmission ABMs from the previous sections it became apparent that each model can take hours to build. It is also known that of all 268 stations only 124 of them have details of station layouts provided by TfL [60]. Because of this, an equation was developed to assess risk at every station. In this section, the focus is to assign values of risk to all stations according to the passenger facilities they possess as well as the pre-established

busyness values. To tackle this problem, empirical risk characterisation was implemented. These values go on to be used as a means of network-wide comparison of risk between the stations.

8.1 Parameters

There are a couple of parameters that can be used as factors for risk evaluation. In this part of the report, only facilities available and the level of the stations were considered in the calculation. This is due to limited information on other parameters as well as to provide a simpler initial evaluation of risk for all the stations. The layout of the stations will also not be included in this evaluation because there is a large gap in sizes as well as features of the stations, making it hard to generalise stations and assign risk values accordingly. Therefore, station features like corridors and stairs were not included in the evaluation process.

8.1.1 Facilities

There are a lot of facilities available in the tube stations such as lifts, and escalators. Among those, only a few have their information available on the TfL website [65]. Not all of these facilities were used in the calculation. Some of them like photo booths and cash machines are not essential to the day to day running of the station. Also in the case of a pandemic, these facilities can be closed to the public and the overall functionality of a tube station remains unchanged. After a process of elimination, the facilities and information that were used in the calculation are number of lifts, number of escalators, number of gates (ticket barriers), and number of platforms.

Lifts and escalators are widely used as the major vertical transportation (VT) available in many tube stations. Most tube stations have multi-level platforms, in which both lifts and escalators are vital in helping passengers' movement within a station. Meanwhile, gates are always located at entrances and exits of a station, where passengers' interaction cannot be ignored. For these three facilities, the main mode of transmission was initially assumed to be via fomites. This is because the facilities have multiple touch points where disease transmission can occur. Other than that, platforms are where passengers alight or board a train, which means passengers will congregate in that area. The main mode of transmission for platforms was initially assumed to be airborne.

These data are available in the form of XML file given by TfL on their Open Data Portal [65]. A processed version of these data can be found on Streetcheck [66] for public use. An excel sheet was created to store these data and used in the calculation part of the model.

8.1.2 Above-ground or Below-ground

Not all LU stations are located under the ground. More than half of them are located above the ground [67]. This is important because the long-range aerosol transmission route will not be considered for stations that are located above the ground resulting in a reduced risk value. These above-ground stations were not excluded like any other label such as DLR and London Overground, because they are vital in the process of re-routing passengers as explored in Section 9, to not obstruct the flow of the train lines.

The information on the stations' location is taken from the map released by TfL that shows which stations are either under the ground or inside a tunnel. This map is mainly used to help people with claustrophobia or anxiety to navigate the subway more comfortably [68]. Stations that are under a tunnel were also considered to be underground in the calculation due to their nature of being in an enclosed system, hence long-range disease transmission cannot be ignored.

8.2 Risk Equation

Based on the literature review conducted, Equation 12 is formed as a basic structure for categorising risk. The resulting risk value, R_s is a relative value assigned to each station for comparison purposes. A general movement of passengers was considered as shown in Figure 18 below. The starting point is assumed to be the entrance of the station, where gates are located while the ending point is the platforms of the station. Connecting the two points is the vertical transportation (VT) available in each station.

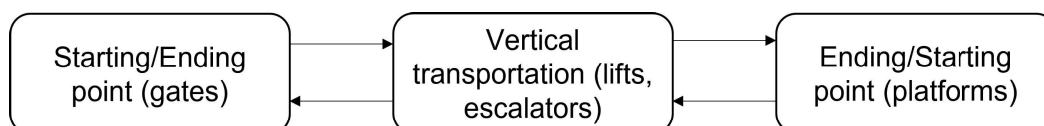


Figure 18: General passengers flow in a station.

Passengers will flow from the starting point to the ending point while using the VT available. Initial risk is calculated using Equation 12:

$$R_s = N \left[\left(R_a \frac{1}{P} \right) + R_f \left(\frac{\alpha}{L} + \frac{\beta}{E} + \frac{\gamma}{G} \right) \right], \quad (12)$$

where N , P , L , E , G , are the number of passengers, platforms, lifts, escalators, and gates (ticket barriers) for every tube station respectively, R_a and R_f are the relative importance of airborne and fomite transmission respectively, and α , β , and γ are the probability of infection for lift, escalator, and gate respectively.

8.2.1 Percentage Usage

Not all the facilities are used by the passengers in the tube stations. In the case of stations with lifts and escalators available, only one of the vertical transportation methods will be used by the passengers to travel from the entrance to the platform. By introducing a set value for different scenarios, the passengers can be divided according to the relative usage of the facilities. These values are then multiplied with each term for each facility in Equation 12.

The first scenario is when there are both escalators and lifts in the stations, the usage value for lifts and escalators are 10% and 90% respectively. These values are due to the assumption that all of the passengers will be using escalators instead of lifts, except for those who have disabilities and carriage [69, 70]. If the stations do not have any of those facilities, the value, as well as the term associated with both lifts and escalators in Equation 12, will be zero. Next, in the case of stations with no lifts but have escalators, the percentage usage for escalators and lifts would be 100% and 0% respectively, therefore the term associated with lifts in the Equation 12 will be zero. All the passengers were assumed to be using the escalators even in the presence of stairs.

The same assumption can also be made when the stations have lifts but no escalators, but the initial assessment shows that some stations are not deep enough for all the passengers to use the lifts, this causes the stations to have high-risk values. To reduce the value of percentage usage for lift, the platform depths for every station that have lifts only were taken into account. According to literature conducted by Swinarski et al. [71], the stairs usage can be shown in Table 1 below. These data were used together with the platform depths [72] to decide the percentage usage for the lifts. If the platform depth of a station is inside the range of the floors travelled, the corresponding value of lift usage will be assigned to that station. For stations with multiple platforms, the average value for the platform depths was used. For gates, the percentage usage, z will always be 100% if said facility is present because all passengers will have to pass through the gates to get in and out of the station, except for stations with no gates where the value of z will be 0%.

Table 1: Values for lift usage will be assigned to stations with lifts only depending on their platform depths [71].

Floors Travelled	Vertical Height (m)	Stairs Usage Up (%)	Stairs Usage Down (%)	Average Stairs Usage (%)	Lift Usage, x (%)
1	3.3	80	90	85	15
2	6.6	50	80	65	35
3	9.89	20	50	35	65
4	13.2	10	20	15	85
5	16.5	5	5	5	95
6	19.8	0	0	0	100

8.2.2 Busyness of the Station

In the Equation 12, the size of the stations is not considered in the calculation. In order to solve this problem, busyness B_s (Equation 2) was integrated into the Equation 12 taken from Section 3.2. This ensure that the crowding of the platforms at the station is well represented.

8.2.3 Risk Coefficients (α , β , and γ)

In Section 6, the tube station is broken down and modelled in different parts. Using several agent-based mini models of the facilities, a simulation was done for each facility to give how disease infectivity

changes with respect to the number of people using the facility. These mini models were used to mimic the situation around the facilities in the station. For gates and escalators, the short-range transmission was considered as well as fomites because queuing was allowed causing contact between passengers, whereas long-range transmission was added for lifts due to low airflow. The number of passengers arriving for every 15-minute time block was then varied from smallest to largest expected flows while the disease transmission parameters were kept constant. The probability of infection was recorded and risk coefficient terms for each facility were formulated via fitting of lines. These results were then integrated into Equation 12 to form Equation 13:

$$R_s = (rB_s) + (A + B + C) \quad (13)$$

where

$$A = 0.0002 \left(\frac{Nx}{15L} \right) \quad (14) \quad B = 0.0004 \left(\frac{Ny}{15E} \right) \quad (15) \quad C = 0.00009 \left(\frac{Nz}{15G} \right)^{0.053} \quad (16)$$

and r is the probability of infection scaled for busyness, given as 0.00196 for below-ground stations, and 0.00098 for above-ground stations. The r value for underground stations was taken directly from the mini model. For above-ground stations, r has halved the value based on the relative contribution of disease transmission done by Azimi et al. [12]. In the case of stations with none of the facilities, the term A , B , and C are equal to 0. R_a and R_f were removed because the mode of transmission was already included in the mini models. 15 in Equation 14 is due to the assumption that a lift can arrive every minute, whereas 15 in Equation 15 and 16 is to find the number of passengers per minute.

8.3 Station Risk Characterisation

A MATLAB script was created to calculate the risk value, R_s for all the stations [50]. The pseudo-code is shown in the Algorithm 1. The data for the number of passengers were taken from the 2019 NUMBAT data [47]. The number of entries and exits for every station was used to represent the number of passengers to calculate the risk value for lifts, escalators, and gates. The number of interchanges is considered in the part of Equation 13 which assesses risk due to busyness however, was not considered in the part of the calculation considering facilities due to the assumption that passengers will not be using these facilities when they change lines or trains. The risk value for each station was calculated starting at 5 a.m to 5 a.m the next day, across a total of 96 time blocks. The risk values are then averaged, to have the idea of the risk present at the station in a day. These results are shown in Figures 23a in Section 10.

Algorithm 1 Pseudo-code for Risk Value, R_s Calculation

Input: Density for each time block for all stations DT , Number of lifts L , Number of escalators E , Number of gates G

Output: Risk for each time block for all stations RT

```

1: for All 268 stations do
2:   Based on the number of facilities, assign the percentage usage of the facilities  $x$ ,  $y$ , and  $z$ .
3:   for All 96 time blocks do
4:     if Any of the facilities,  $L = 0$ ,  $E = 0$ , and  $G = 0$  then
5:        $A$ ,  $B$ ,  $C$  will equal to 0
6:     else
7:        $A$  = Equation 14,  $B$  = Equation 15, and  $C$  = Equation 16
8:     end if
9:     Calculate the value of risk,  $RT$  using equation 13
10:   end for
11: end for

```

8.3.1 Sensitivity Analysis

A sensitivity analysis of the model was conducted to see the effect of changing the parameters on the overall results. The number of all of the stated facilities was varied from 1 to 20, and the percentage difference in the risk value relative to the value with only one facility was calculated. As seen in Figure 19a, increasing the number of escalators will have a greater effect on the overall results followed by

lifts and then gates. This is because escalators were used considerably more than the other facilities in most cases as outlined in Section 8.2.1. Figure 19b shows the effect on the overall risk value when the busyness, B_s was varied across the time block, R_s . The population density of Victoria LU was taken and sorted from low to high. The percentage increase of the density from the lowest value was plotted against the percentage difference in the risk value. It shows a linear relationship because the busyness is directly proportional to the risk value as seen in Equation 13, so long as the number of facilities is constant throughout the time blocks.

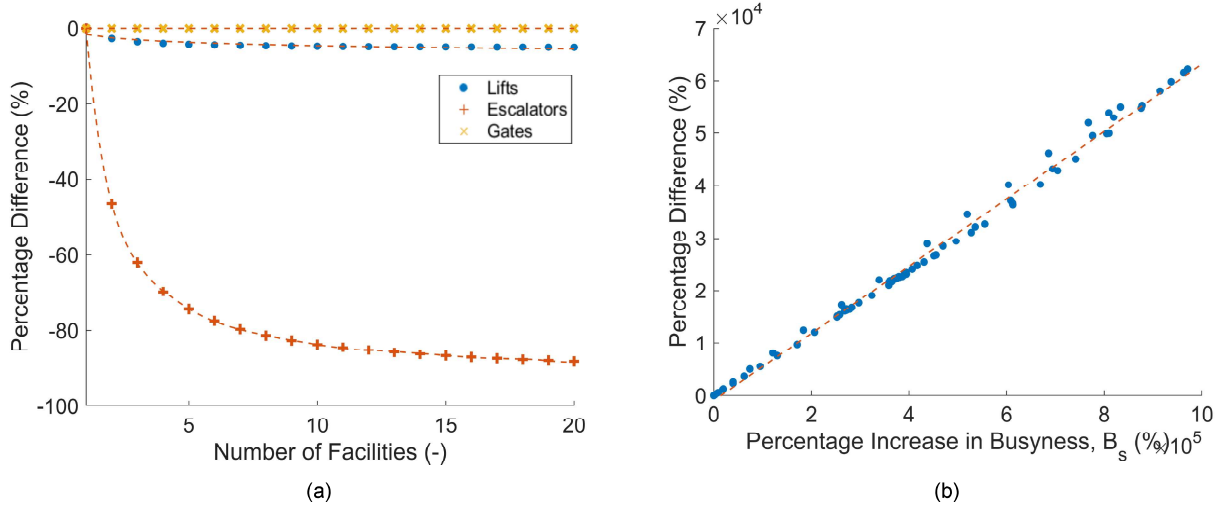


Figure 19: Sensitivity analysis of the equation. Figure 19a shows the effect of changing the number of facilities to overall results. Figure 19b shows the effect of increasing value of busyness, B_s to the risk value, R_s .

9 Routing Passengers to Avoid Risk (OP)

One of the objectives of this study is to assess and suggest means by which passengers can reduce risk to themselves. One solution is to reroute passengers so as to minimise their likelihood of infection whilst travelling. This section will consider the infection risk as that of COVID-19. A dynamic routing algorithm has been devised using pre-pandemic 2019 NUMBAT data [47], with the intention of minimising passenger risk as an objective function.

9.1 Assessing Infection Risk

In order to minimise a passenger's likelihood of infection it was first necessary to devise a means of quantifying it for passengers in both the train and the station environments. Empirical risk values for the station have already been established in Section 8 of this report and describe the risk to a passenger when exposed to an individual station environment for 5 minutes.

In order to model the risk of the train environment a small situational ABM was developed to simulate the risk of transmission with respect to busyness and time of exposure within a virtual box which is dimensionally representative of a Bakerloo line MK1 and MK2 tube stock train carriage [73].

From Section 5 it has already been established that in accordance to the ABM the probability of COVID-19 infection can be assumed to increase in a linear fashion with respect to time of exposure. With this in mind a series of busyness variation experiments were conducted in order to deduce a trend for the effect of busyness on COVID-19 transmission within the train carriage environment. The experiments involved altering the number of passengers within the carriage, in a constant time scenario, in order to deduce the relationship between busyness and the probability of infection. The final relationship between risk of infection, the number of passengers and the exposure time in the train carriage environment is shown in Equation 17:

$$R_c^{XY} = kN_c^{XY}t_{XY} \quad (17)$$

where R_c^{XY} is the risk of infection in the carriage environment between stations X and Y , N_c^{XY} is the number of passengers on the carriage between stations X and Y as given in the NUMBAT 2019 data [47], t_{XY} is the time of passenger exposure between stations X and Y and k is a co-efficient equal to 4×10^{-4} relating the risk of infection to the terms for number of passengers and time of exposure.

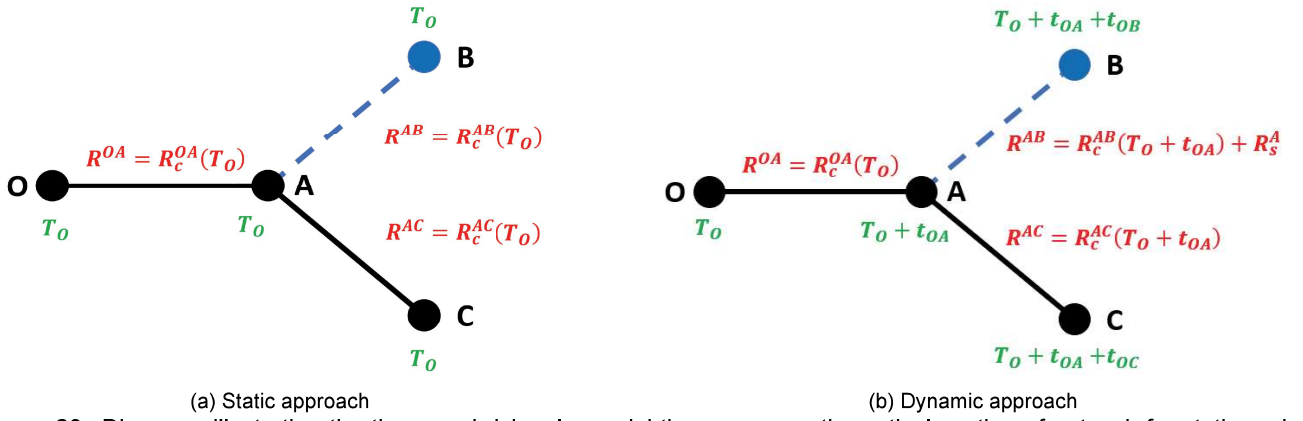


Figure 20: Diagrams illustrating the times and risk value weightings across a theoretical section of network for static and dynamic routing approaches. Figure 20a shows the static approach where O , A , B and C represent different stations with O representing the starting station, R represents the total risk value between two stations given in superscript, R_c is the risk of infection in the carriage environment between two stations given in superscript (see Equation 17) and T_0 represents the starting time of the journey. Figure 20b shows the dynamic approach where t represents the time on a train between stations given in subscript and R_s gives the risk from exposure at a station given in superscript for five minutes. The blue dashed line symbolises the event that a passenger changes line.

9.2 The Routing Algorithm

When selecting a routing algorithm two different options were considered. The first is the well known Dijkstra's algorithm [74]. Dijkstra's algorithm is a K-search algorithm which works to find the shortest path between the assigned origin and destination nodes in a graph [75]. The algorithm systematically searches neighbouring nodes which correspond to the path of the lowest weighting. The most common variant of Dijkstra's algorithm fixes a 'source' node and finds the shortest path to all other nodes in the graph, hence producing a shortest path tree [76]. Unfortunately, the fact that the algorithm visits every node in the network means that running the algorithm has a high time complexity and demands a large storage capacity [77]. Further to this the single source shortest path problem addressed in Dijkstra's algorithm lacks the ability to computationally process parallel solutions, hence prohibiting against the use of computational techniques aimed at decreasing the run-time of an algorithm [78].

The A^* algorithm is an adaptation of Dijkstra's algorithm, which employs the use of a heuristic measure (often euclidean distance) to make it into an informed search algorithm [79]. This has the effect of considerably reducing the time complexity of the algorithm as it no longer has to visit every node in the graph. However, for this algorithm to be effective it is important to establish a reliable heuristic value, which is representative of the path the algorithm will follow. If the heuristic is incorrect or unreliable then the algorithm can output an incorrect value for the shortest path between the origin and the destination.

From these two algorithms, the Dijkstra's algorithm was chosen for implementation to the route planner. It was decided that given the link weightings chosen for processing by the algorithm are representative of risk, creating a heuristic value of risk for use in the A^* algorithm would be problematic and unreliable, potentially resulting in an incorrect result for the shortest path. Further to this the time complexity and large storage capacity of Dijkstra's algorithm are less important when considering the scope of this project and given that the network is only comprised of 268 nodes, the time implications of using Dijkstra's algorithm are manageable.

The complexity in the routing algorithm lies with its ability to function dynamically. Two possible methods were available for routing to avoid risk: static routing and dynamic routing. A comparison of the static and dynamic approaches to routing passengers by risk in the TfL network is illustrated in Figure 20. In contrast to the simpler static routing approach, dynamic routing accounts for perturbations and changes which occur while the journey takes place. In the route planner developed for this study two significant dynamic features were involved. The first is the inclusion of the time taken to traverse the network to a node in subsequent risk calculations as shown in Figure 20b where the dynamic risk values are shown to be a function of the cumulative time to reach that point. The second is the implementation of a station dependent risk penalty incurred if the route requires line changes. This is shown in Figure 20b where the risk term for the blue dashed line (signifying a line change) involves an R_s risk term incurred from pausing at the station. Both of these changes are impossible for the simpler static routing approach to account for as it assumes time to be constant throughout the journey. This

significantly hinders its ability to represent passenger risk. Especially when considering long distance journeys because as shown in Section 3 network busyness varies across the course of the day.

Algorithm 2 describes the function of the algorithm in the form of pseudo-code. In the pseudo code an open node describes a node which has not been visited by the algorithm and is valid for processing in the shortest path calculations and a closed node describes a node to which all valid routes have been discovered and the shortest one established, meaning the node has been visited rendering further calculations on this node unnecessary.

Algorithm 2 Dynamic Dijkstra Implementation

```

1: Let O = Origin Node, D = Destination Node, C = the Current Node, S = Surrounding Nodes, R = Risk of
   Journey and t = time.
2: Assign risk of O, R = 0.
3: Set time t = 0 and C = O.
4: Initialise list of open nodes and closed nodes
5: while C ≠ D do
6:   for All S of C (S cannot be a closed node) do
7:     if The individual node of S is not in the list of open or closed nodes. then
8:       Add the node of S to the open list of nodes
9:       Calculate the risk (R)  $O \rightarrow C$ , ensuring to account for the time t taken to reach C.
10:      if  $C \rightarrow S$  is on different TfL line to previous link then
11:         $R = R + \text{risk associated with exposure to station at node C for 5 minutes.}$ 
12:      end if
13:      if  $R < \text{existing R value for the selected node}$  or if there is no existing R value for the selected node
then
14:        update R value of selected value of S
15:        Select the parent node of S to be C
16:      end if
17:    end if
18:    Assign the next node of S.
19:  end for
20:  Add C to closed nodes list. When a node becomes closed it can no longer be re-visited by the algorithm.
21:  Remove the node with the lowest R value from the open list and assign it to current node C.
22:  Update the time t so that it accounts for the journey to get to C.
23: end while

```

A series of assumptions were made during the making of the route planner in order to simplify the problem and enable the faster construction of the routing algorithm. The first is that the risk of infection on the trains (Equation 17), derived using an ABM for one type of train, is applicable universally for all trains across the network. Constructing an agent based mini model for every type of train in the network was not possible within the time-frame of the project.

The second assumption made is that the length of time that a person is exposed to the station environment when they change lines is 5 minutes. This value was necessary to create due to a lack of access to a comprehensive database for TfL train timetabling. It uses the average frequency of train arrival across the entire network across a 24 hr period in order to provide a worst case scenario and ensure the avoidance of crowding at high risk stations if at all possible. (see Section 8)[47].

The final assumptions made are that journeys between stations are a straight line between the GPS latitude and longitude positions for each station [80], that all trains travel at the TfL stated average speed of a train [81], and that every train pauses for 30 seconds at each station. The pause at each station is an estimated value due to lack of available data, however it plays a crucial role in ensuring that routes which are geographically shorter distances but contain more stations aren't necessarily quicker.

9.3 Routing Outputs

Two different journeys were set up so as to test the dynamic routing algorithm in different scenarios. The first scenario is between Bank and Monument and Oxford Circus beginning the journey at 5.30pm. This route was chosen as according to the NUMBAT 2019 passenger flow data between Monday and Thursday these two stations boast the greatest number of passengers [47]. 5.30pm was chosen because as shown in the busyness profiles in Section 3.3 the afternoon busyness peak is shown to occur

at this time. This journey therefore can be said to be representative of the interest of a large number of the TfL commuters (see Section 1.2). The second journey investigated is between South Ealing and Epping underground stations. This provides a measure for how effective re-routing passengers is across a long distance journey, while still passing through central London.

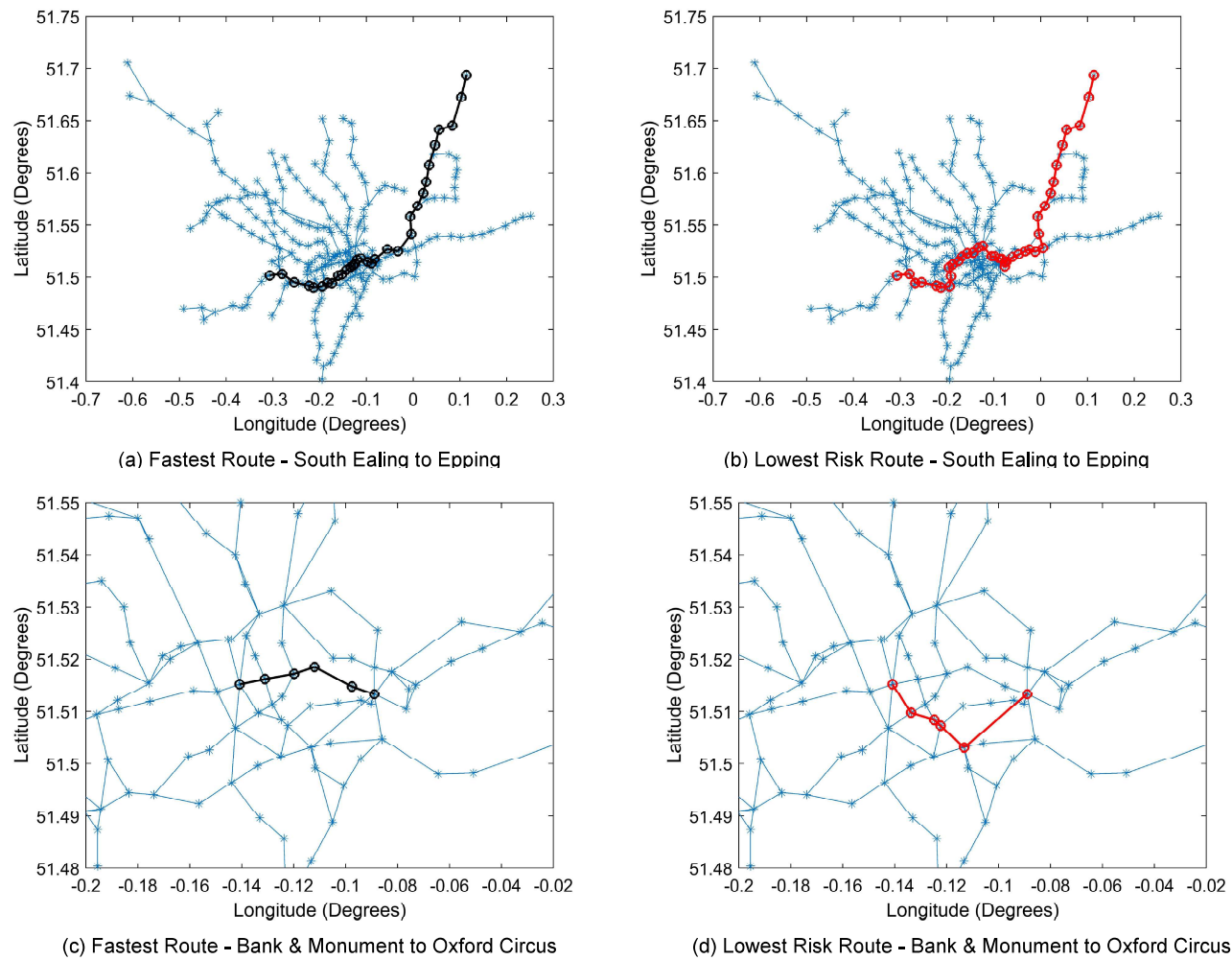


Figure 21: A representation on a map of the TfL network of the routes chosen by the routing algorithm when it is programmed to find the fastest route (black), and when it is programmed to find the lowest risk route (red). Every other station and line in the network is represented in blue.

Table 2: The values for the time taken and the comparative total risk of the journeys illustrated in Figure 21.

Routing Type	From/To	Time Taken (mins)	Comparative Risk Value (-)
Fastest Route	South Ealing - Epping	127	2.70
Lowest Risk	South Ealing - Epping	216	2.05
Fastest Route	Bank & Monument - Oxford Circus	13	0.40
Lowest Risk	Bank & Monument - Oxford Circus	24	0.20

9.4 Verification

In order to verify the use of the routing algorithm for dynamically calculating the shortest route in the simulated TfL network the results for the fastest route calculation are compared to the TfL journey planner available on the TfL website [82].

Table 3 shows that the chosen fastest route is in agreement for both means of calculating the quickest journey. However, for both different journeys a significant disparity in journey time is recorded. Between South Ealing to Epping underground stations the Dynamic Dijkstra's algorithm implemented shows a 49.4% increase on the time taken in the TfL journey route planner. For the Bank and Monument to Oxford Circus route a 62.5% increase is seen.

Table 3: The values for the fastest route journey times and the lines taken from South Ealing to Epping underground stations and from Bank & Monument to Oxford Circus underground stations for the TfL Journey Route Planner [82], and the Dynamic Dijkstra's algorithm applied in this report.

Route Planner	From/To	Time Taken (mins)	Lines Taken (-)
Dynamic Dijkstra's	South Ealing - Epping	127	Piccadilly, Central
TfL Route Planner	South Ealing - Epping	85	Piccadilly, Central
Dynamic Dijkstra's	Bank & Monument - Oxford Circus	13	Central
TfL Route Planner	Bank & Monument - Oxford Circus	8	Central

10 Network-Wide Risk Analysis Toolkit (JH, WA, OP, IJ)

Now that the components have been developed, the full toolkit can now be integrated. This aims to meet all of the objectives outlined in Section 1.2 thus fulfilling the original aim set.

Figure 22 provides a diagrammatic representation of the recommended network-wide analysis approach. This has been split into three key sections. First, network risk is characterised. Second, a particular station on the network is selected and modelled so that policy analysis can be performed and transferred to similar stations. Finally, routing is used to provide passengers with a means to avoid risk. The effect of passenger routing has already been explored in depth in Section 9. This section will therefore focus on the network analysis and station analysis sections of Figure 22.

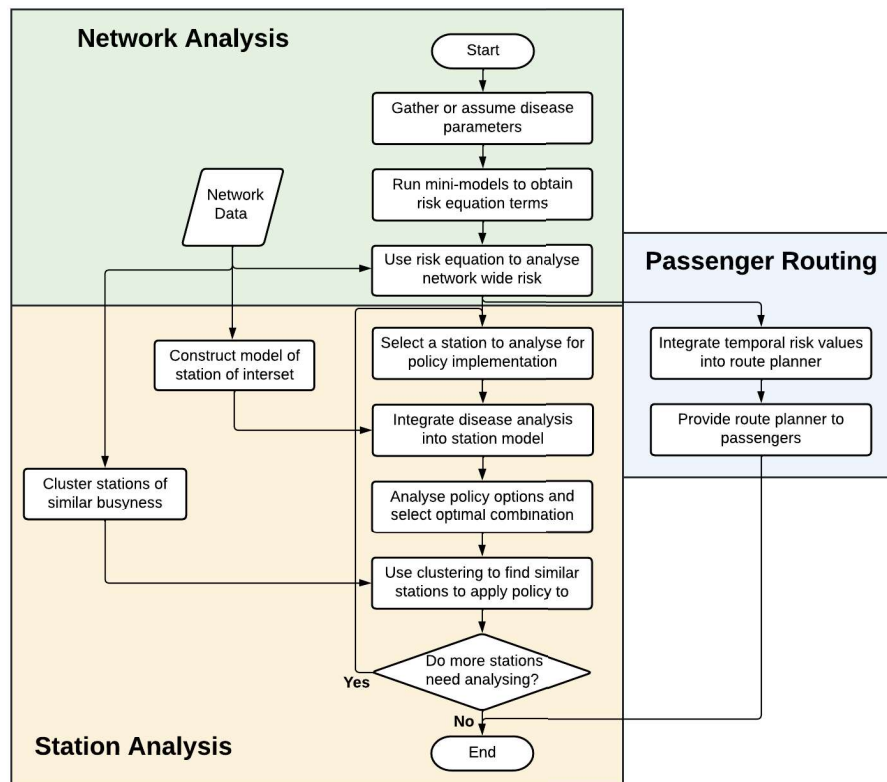


Figure 22: The suggested analysis process with the three key sections highlighted.

10.1 Network-Wide Station Risk Comparison

Performing network wide risk analysis is a three step process of setting disease parameters, obtaining risk equation terms via mini-models, and using the risk equation on network data.

As this report focuses on COVID-19, this was the disease chosen to analyse. Following the pandemic much research has been conducted into the infectivity of the disease providing a multitude of sources for parameters. These were reviewed and selected values and distributions are provided in the Appendix.

Analysis was performed on an early pandemic scenario with a high level of infected individuals. UK case data provides peak numbers of 7.5% [83] infected population at this stage. A value of 10% was chosen to account for a larger infection density in the capital and asymptomatic cases not counted. Additionally, the number of recovered was set to 0 to represent the population having no immunity or

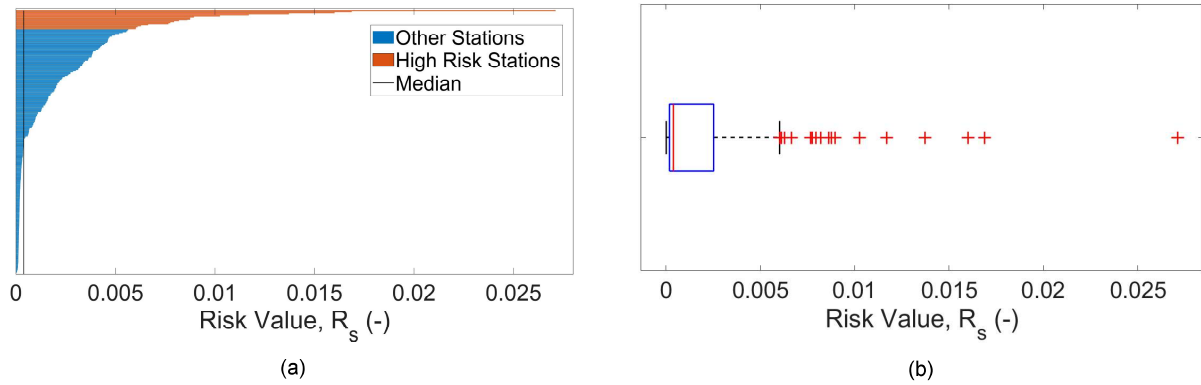


Figure 23: The distribution of risk value, R_s for all stations are shown here. Figure 23a shows the average risk value, R_s for all 268 LU stations. Figure 23b shows the box plot of the risk value, R_s for all the stations.

vaccination.

With disease parameters gathered these were inserted into the mini-models. The models were ran as described in Section 8.2.3. A risk value for each station was calculated as in section 8.3. The averaged risk, R_s , values are shown in Figure 23a and as a box plot in Figure 23b. Several stations have a very distinctive difference in risk value from the other stations, called the outliers in the box plot because they are 3 standard deviation away from the mean risk value, and therefore, these stations are considered to be high-risk stations. All nineteen of these stations are shown in Figure 24.

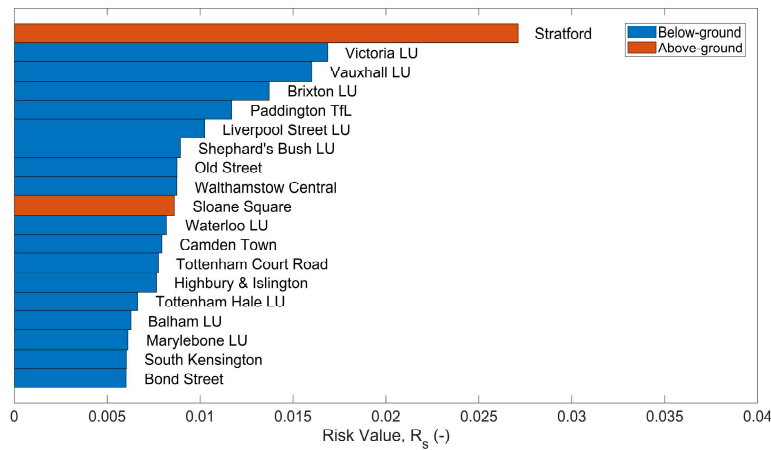


Figure 24: Top 19 Underground Stations with the highest average risk value, R_s .

10.2 Evaluation of Policy Effectiveness

From the ranking of highest risk stations presented in Figure 24, Vauxhall station was selected to perform policy analysis on. This was selected at it was the station with the third highest risk with the two prior being large station complexes which would have large computational times, limiting the analysis that could be performed with the resources available. The following analysis does not aim to present a genuine suggestion for policy, only a proof of concept of this modelling approach.

10.2.1 Integrated Station Model Creation

This station was modelled as outlined in Section 6. The required TD matrices were generated and the comparison verification check performed yielding 98.9% GEH values below three when taking an average across ten runs.

Disease modelling was integrated as described in Section 7 with parameters carried over from the previous section.

10.2.2 Policy Implementation in Vauxhall LU Station

Firstly, the model was ran with zero policy implemented to gain a baseline value. This resulted in a probability of infection of 6.0%. Dose was tracked across this analysis run; this data is displayed in Figures 25 and 26a for short range and fomite transmission respectively.

From Figure 25 problem areas can be selected around the entrance ways where there are bi-directional flows (A), along the escalators (B) and in the lower hall at the base of the escalators (C). For the first of

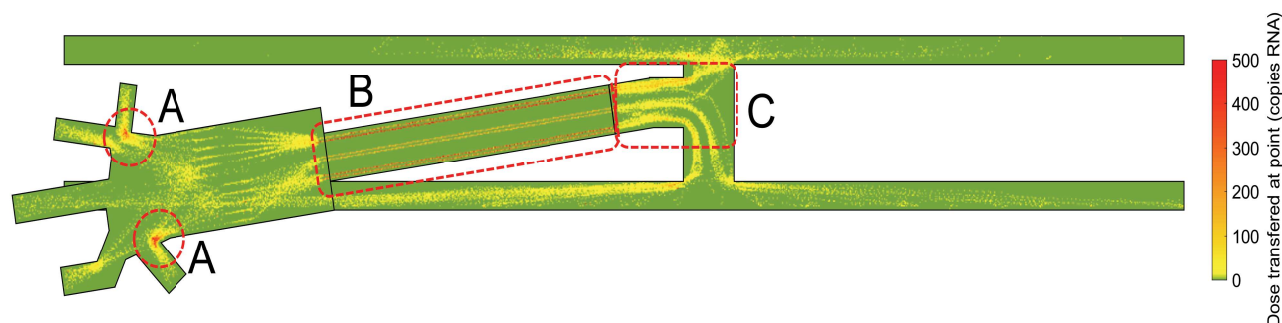


Figure 25: Heat map of dose transferred in the station via short range transmission before policy was modelled. A, B and C are selected areas of high risk for which policy is targeted.

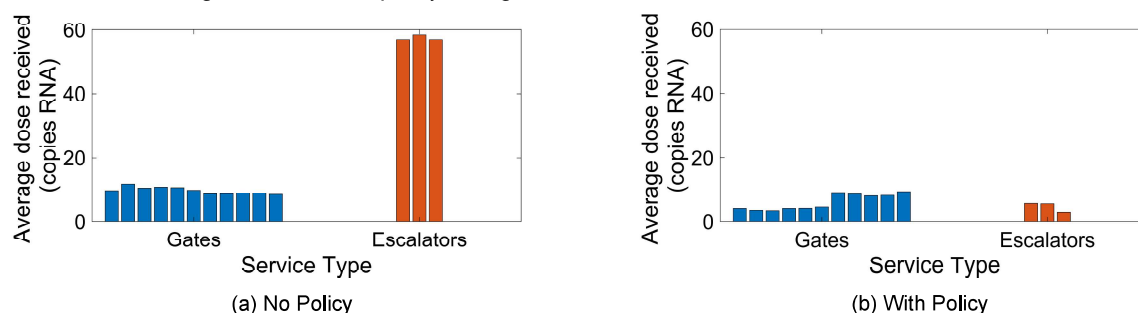


Figure 26: Average dose received by a susceptible agent when using each service both without policy and with policy implemented.

these issues a one-way system was proposed with three of the entry ways dedicated for those leaving the station and the remaining two for those entering. Due to the relative tightness of the lower hall and lack of alternative routes, similar policy could not be adopted for second and third areas. For targeting this area, and as general station coverage, a mandatory face mask policy is suggested alongside social distancing. However, social distancing could not be modelled, an issue later discussed in Section 11.

From Figure 26a it can be seen that escalators have a larger dose transferal than gates. Fomite transmission accounted for 83% of total dose transferred so is currently the dominant transmission route. Two policies were implemented: regular cleaning with each touch point being cleaned every 15 minutes; and hand-sanitiser stations positioned in the entrance hall area to promote hand washing before fully entering and when leaving the station.

Each of these policies was modelled and tested in turn then tested with all policies implemented. The new probability of infection when adopting each policy is tabulated in Table 4 along with the percentage improvement over the station with no policy.

Table 4: Change in probability of infection as policy options applied.

Policy Applied	Estimated Probability of Infection (%)	Decrease in Probability of Infection (%)
No Policy	6.0	-
One-Way Only	5.9	1
Masks Only	5.2	13
Cleaning Only	2.7	55
Hand Washing Only	4.4	27
All Previous Policy	1.2	80

With all policy implemented the probability of infection drops to 1.2%, a decrease of 80% over the original probability of infection. Cleaning has the largest impact on risk reduction followed by hand washing. These combined reduced total fomite transmission by 85%. Figure 26b shows the average dose received for each service now including policy. Comparing with Figure 26a it can be seen that this has had the greatest effect on escalators. It has some effect on gates but not to the same extent; this will be due to the effect of the surface areas. As escalators have a larger surface area dose takes longer to build up so the cleaning cycle means that build up is minimal. Gates have a lower service area so build up is quicker and returns to the same level of risk. Four gates remain at a similar level of

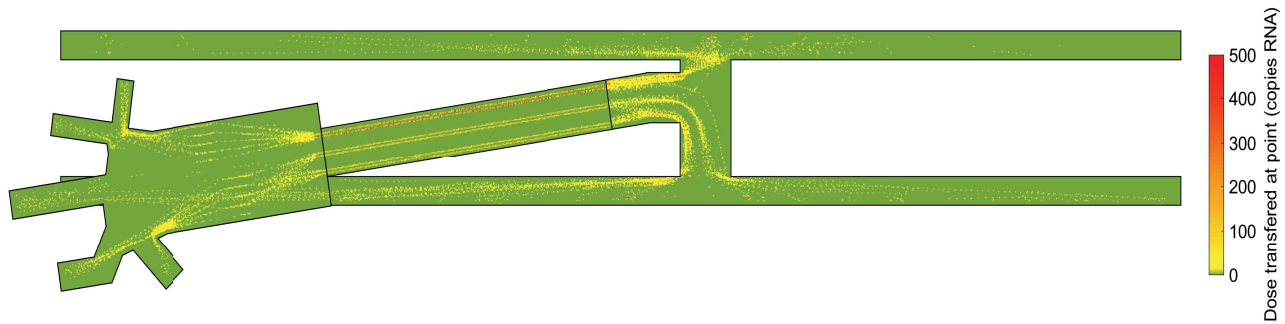


Figure 27: Heat map of dose transferred in the station via short range transmission with policy modelled.

transmission as these are the gates for passengers leaving the station, hence these individuals are not effected by the hand washing policy.

The short-range-targeted policy has had less effect. This is due to the station remaining crowded in many areas resulting in social distancing being impossible. Dose was again tracked and shown in Figure 27. It can be seen that the one-way system did reduce dose transfer in the entrance area but this had minimal effect. This is likely due to the 360° nature of the contacts at this stage, an issue discussed later. The escalators and lower hall remain at a similar level due to the lack of social distancing.

10.2.3 Network-Wide Effect of Policy Implementation

The network-wide effects of policy implementation are calculated by applying the tabulated probabilities of infection (Table 4) to all similar stations. A similar station is defined as a station which shares the same station features (excluding number of gates), such as number of platforms, escalators etc. A similar station also has to have been assigned to the same group via clustering of busyness profiles as seen in Section 3.3. The busyness profiles and station features act to influence many of the ABM simulation properties. Examples include passenger behaviour and flow dynamics, the number of interactions experienced by passengers and the effectiveness of cleaning schedules due to infectious dose accumulation.

From all the 268 stations in the network, 14 were found to share similar station features and profiles for busyness. To account for the difference in passenger density at each station, the mean busyness of each station is taken to linearly scale the probability of infection with respect to the mean busyness value at Vauxhall station. The assumption that probability of infection scales linearly with busyness has previously been discussed in Section 8 of the report.

Table 5: The rate of infection and the average probability of infection across the 15 stations as policy options applied. The total number of passengers passing through all 15 stations in one day is 493352.

Policy Applied	Number Infected (per day)	Average Probability of Infection (%)
No Policy	15000	3.1
One-Way Only	15000	3.1
Masks Only	13000	2.7
Cleaning Only	7000	1.4
Hand Washing Only	11000	2.3
All Previous Policy	3000	0.6

11 Discussion and Further Work (JH, WA, OP, IJ)

Due to the multi-faceted nature of this study the discussion will be structured so as to discuss the individual methods used in building the tool explored in Section 10.

11.1 Analysing Network Risk

Empirical risk calculation essentially aims to rank the stations according to the risk of disease transmission. Stratford is found to be the station with the highest value of risk averaging at 2.711×10^{-2} , whereas Roding Valley has the lowest risk value averaging at 1.418×10^{-5} . The median risk value was found to be 3.941×10^{-4} as shown as a vertical black line in Figure 23a. Nineteen stations are isolated as the highest-risk stations in the LU network. Among these, there are only two stations that are located above the ground, while the remaining seventeen are underground. Stratford was valued

the highest due to it being one of the busiest stations but having fewer facilities available. Meanwhile, Roding Valley has the least average number of passengers in a day, and it is also located above the ground with no facilities available at all [47].

There are some limitations to this method of calculating risk. Firstly, the location of the facilities in the layout is not considered. It can be said that usage of the facilities scales with the convenience of use. Given Finsbury Park for example, the lifts are only accessible if the passengers get into the station using the National Rail entrance, whereas passengers who use the Underground entrance will not be able to use the lifts.

Other than that, the number of passengers using the facilities was assumed to be equally divided among the number of facilities. This assumption does not reflect the reality of human behaviour in the stations. Usually people will use whichever facilities are nearest to them or show the least occupancy. Furthermore, there might be some facilities being used more than the others during peak busyness time in certain stations. For instance, four times more people are exiting the Canary Wharf station compared to people entering the station in the morning peak [47]. This will cause the escalators heading up towards the exit to be considerably more crowded than in the opposing direction. This would increase the risk value of that station. This model can be further improved by separating the facilities according to the direction of passenger flow.

11.2 Disease Transmission Modelling

The second objective in the completion of the project aim was the development of a disease transmission model. This made use of the exponential dose response model applied with an agent based approach considering three key transmission routes for respiratory diseases. The model also includes other dose response relationships so a range of infectious disease could be effectively modelled. This approach means that the disease transmission model can quickly adopt parameters to mimic new emerging pandemics, resulting in faster analysis and intervention. This is advantageous for passenger safety and ensures public transport remains commercially viable, hence acting in the interest of network executives. Sensitivity analysis was performed to provide an insight on which disease parameters have the largest impact. This aids future modellers by focusing their efforts on the most important parameters.

Whilst the proxy validation of the model was within accepted bounds, the output risk values from the super spreader event validation revealed uncertainty and potential error between 12% and 66%. The uncertainty presented in the risk values suggest that the model is appropriate for comparative use but requires further work before the risk values can hold confidence. This can be rectified through the acquisition of more case data at the same scale as the model and over similar time scales.

Assumptions were made in the creation of the model. The first is that AnyLogic's network feature is applied in all directions around the agent and as such droplets and short-range aerosols are assumed to be emitted in all directions. This fails to account for the geometry of interactions and the directional bias of exhalation and symptomatic behaviours. This likely increases the infection risk presented in scenarios with large crowds and undervalues mitigation strategies such as one way systems. Further work to include this would require lowering the level of abstraction and including a directional conical droplet zone.

The well mixed assumption is used for long range transmission. This limits the mechanism's validity to smaller volumes. The dimensions of stations within the LU are not publicly available and as such the limit of validity with respect to particular stations is left to the user of the tool. Whilst this could be solved via the application of CFD, the computational costs associated would also scale. Surrogate modelling techniques, where high risk areas (low ventilation rates, small volumes and high footfall), are separately modelled and applied could be adopted to reduce the computational strain.

Currently the model does not consider agent behaviour beyond the agents departure from the station. Due to the relatively long half-life of the virus on human skin, the dosage present on an agents hand still presents appreciable risk beyond the station. Fomites are also considered to only exist on the agents hands and fomite agents, but in reality deposition from aerosols and droplets results in agents' apparel becoming fomite carriers, leading to increased risk.

11.3 Integration of Disease Modelling with Station Modelling

The third objective set was the integration of disease modelling with modelling of a station. Modelling stations featured the culmination of pedestrian dynamics and DES station modelling. A framework for

producing station models was produced with the aim of making the modelling of any station on the network possible. Disease modelling was then integrated through the use of insertable agents.

Station modelling was successful, providing a suitable selection of station components which could be used to model a particular problem. This could be verified but validation was limited by the lack of additional data. This also limited the level of abstraction that could be achieved and many assumptions had to be made limiting confidence in the applicability of the results obtained. However, additional datasets could ratify these issues quickly.

Additionally, potential instability issues were discovered when attempting to validate pedestrian crowd behaviour. This issue was not explored due to time constraints. All models utilised in this report had flows below the instability density, though robustly modelling busier stations would require exploration into this issue.

Finally, scalability issues led to computational difficulty on larger station complexes. This could be solved by dedicating more computing power to the issue in the short term. Over a longer timescale other software's could be investigated with better scalability. Provided an agent-based approach is still taken, there have been no limitations found that would prevent disease modelling being transferred to this new software.

Integration of disease modelling has been comprehensively proven to be a viable process for analysing disease transmission in public environments. The agent-based approach makes this viable as a fast integration process and one that can be performed on a pre-existing station model. This is beneficial for modellers as it means that station risk could be found quickly using ABM so that more comprehensive analysis can be performed on a larger set of stations. The time saved means the passengers can be protected quicker and saves money, both advantageous for network executives and the passengers themselves.

11.4 Analysing Policy Effectiveness

Through the use of the toolkit a high risk station was selected and used as a case study for policy analysis, competing objective five. Reduction of probability of infection provided insight into the most effective policy which could be selected. It was estimated that only regular touch-point cleaning produced a 55% reduction in perceived risk while a set of four policy options produced an 80% reduction. This is vital information that station and network managers can use to optimally select the best policy options for one or a selection of stations.

A consideration not explored was how these options can be offset by their costs, both monetary and social. Offsetting by cost would give insight into the best options if there is a limited budget or strict policy was likely to lead to a reduction in ridership and, therefore, revenue.

Confidence in the values provided by policy analysis are subject to the same concerns outlined in Section 11.2. Of particular issue is non-directional interactions limiting one-way system policy as discovered in Section 10.2. Pedestrian modelling also limited policy implementation. Robust modelling of social distancing was not achieved. This was due to many of the parameters that define the social force model being inaccessible to the user in the AnyLogic pedestrian library. While there is a built in social distancing function, this is far too simple for large stations as agents do not know when to break social distancing to ensure flow is not obstructed. A workaround could likely be performed but time constraints prevented further research into this.

Developing solutions to these problems are short term goals. In the long term this analysis tool could be expanded to consider additional forms of public transport. This could come in the form of a pre-integrated modelling package for generalised use cases.

11.5 Network Data Analysis

By analysing network data it is possible to establish patterns and trends which can be studied in order to improve network wide disease mitigation strategies and can increase the range of application for the findings from the ABM. This meets objective 5. This acts in the interests of Network Executives by providing them with a prediction for how effective network policy is likely to be and Network Passengers by reducing their risk of infection.

A means of quantifying busyness was presented which illustrates the distribution of crowding in the TfL network at peak times. The results illustrate that busyness and therefore risk of infection in stations

increases towards central London and shows peaks almost universally across the network at both 8.30am and 5.30pm. This is due to the concentration of commuter flow into central London.

A clustering algorithm was then applied to group stations with similar busyness profiles. The intention was to investigate the scheduling of sanitation rotas to synchronise with busyness profiles. Due to time constraints on the project the effects of altering the cleaning schedule in such a fashion could not be investigated in the ABM. This can be the subject of further work.

This network wide analysis has several restrictions on its validity. Firstly, the busyness definition utilised proxy values. Number of platforms and train capacities at each station are used as representative values for the floor area occupied by crowds. This is a limitation as every station is topographically different. The busyness in a station would be influenced by a number of factors other than the proxies chosen. Further work should seek to acquire more comprehensive details for the floor plans in each station to improve confidence in the busyness metric.

Secondly, the busyness profiles for stations are clustered based on data from only Monday through to Thursday from one week in the year of 2019 [47]. Stations experience varying seasonal and day to day trends and are therefore likely to be grouped differently. Furthermore, investigating human behaviour in the midst of the pandemic, as is available in the NUMBAT 2020 dataset, would gain an insight as to how busyness dynamics change as the pandemic progresses and restrictions are imposed. As this extends beyond the time constraints of this project it is another area for further research to develop the busyness clustering method to group stations using more varied data.

Improvements could be made to the clustering method employed. Firstly clustering by similarity in maxima and minima locations does not account for their magnitude. This means that small scale maxima and minima can result in the creation of new station groupings, or can associate two stations with different scales of busyness but the same peaks. Examples of this can be seen in Figure 5b. Similarly, this approach means gradients of incline and decline for the busyness profiles are neglected, resulting in large differences in grouped stations profiles. Using the same example of Figure 5b, with all stations scaled such that their average busyness is equal, Charing Cross and Piccadilly Circus show a maximum difference for the grouping of 64.5%. This difference is significant and can have repercussions on the validity of the assumption that policy can transfer between stations within the same grouping. Further development could look to link busyness similarity through other means.

11.6 Routing Passengers to Avoid Risk

The dynamic routing algorithm aims to minimise risk to the public by means of optimising the route taken by a passenger as a solution to objective six. This serves all stakeholders in a network service provider role as well as customers by providing a safer working and commuting environment. Table 2 suggests the potential for a 24% and 50% decrease in risk of infection for a long haul journey a short distance journey respectively when compared to the fastest route. It is clear that for both stations and lines the route planner demonstrates the ability to relieve pressure on high risk areas of the network.

The low risk routing algorithm has severe journey time implications. Tables 2 and 3 suggest that a low risk journey could take up to 2 hrs 11 mins longer than its TfL journey planner counterpart. This represents a large time penalty given that for many passengers achieving the shortest commuting time is a priority. Section 9.4 highlights how this could have been an overestimation of journey time. The Dynamic Dijkstra's routing algorithm showed a 49.4% and a 62.5% increase on the time predicted by the TfL route planner for South Ealing to Epping and Bank & Monument to Oxford Circus respectively. The inconsistency highlights a major limitation. Because the time difference relative to the TfL route planner varies dependent on the journey taken, it can be concluded that the assumptions made due to lack of available data mean that the route planner is untrue to reality. This means that the modelled exposure times for passengers are inaccurate and consequently so are the assigned risk. This could be remedied with data true to the TfL network. Additionally further testing should be performed to comprehensively gauge the network wide effectiveness of routing in this fashion.

Careful consideration would be required prior to implementing the route planner. Use in the public domain could take the form of a mobile application or web based user interface. An incentive might be considered to compensate for the inconvenience of the extra duration of the journeys. This could take the form of discounted journeys. Furthermore, prior to implementing the journey planner, development of the routing engine would be required. The assumption that an individual is travelling during the NUMBAT timescales provided means the route planner cannot utilise busyness forecasting or predictions.

It also cannot consider the effect it itself has on busyness in the network. Further work could integrate a real time predictive feature to the routing engine, enabling it to predict busyness and epidemiological trends. Tools used in such applications include dynamic linear modelling and seasonal auto-regressive integrated moving averages (SARIMA) [84]. Furthermore, the computational complexity of the dynamic Dijkstra's algorithm make it unsuitable for practical use in an application or online. Computational efficiency improvement could be made using methods for improvement from literature include restricting the search area for Dijkstra's algorithm, or using heuristics [85]. Lastly, to further improve the routing algorithm a compromise with other criteria deemed important for passenger journey satisfaction should be made, such as rider patience and comfort.

12 Conclusions

The aim of this project was to provide a toolkit for analysing disease spread in urban underground rail stations so that decisions on policy can be informed and effective. The implementation of the tool for Vauxhall and all similar stations predicts that via the application of one way systems, the wearing of masks, enhanced cleaning procedures and regular hand washing approximately 12200 less people will become infected each day network wide. For Vauxhall and all stations defined as similar this corresponds to a decrease of 2.5% in a given passengers risk of infection. Further to this, the applied routing tool demonstrated the ability to reduce journey risk by up to 50% relative to the fastest route of travel. This toolkit can therefore be seen to provide TfL with a means of anticipating the effects of implementing mitigation policies and therefore assessing their commercial validity and epidemiological benefits across the network.

In the long term it is anticipated that epidemics similar to COVID-19 will become more frequent and harder to contain. Given further development, the toolkit produced in this study can act to save thousands of lives through the means of effectively quantifying the impact different disease mitigation strategies can have. By doing so, decision makers are empowered with the knowledge needed to correctly select restrictions required to limit the spread of disease. To maximise the toolkit's effectiveness real time data on passenger flows and TfL details on workforce capacity and capabilities should be integrated to the tools presented hence producing a live feedback mechanism for minimising infectivity on the underground. It is also important to emphasise that the toolkit would be transferable to most large scale public transport networks including flights, overground trains and buses.

Overall this study has illustrated how through the holistic analysis of the TfL network a toolkit can be devised to quantify the effectiveness of policy implementation. Through the use of locally applied agent-based modelling, contagiousness of disease in TfL underground station environments was analysed to a precision of 15 minutes. This was expanded to a global network scale through the application of risk analysis and routing techniques. Further work should decrease the level of abstraction in all techniques employed and more comprehensively validate the toolkit, providing greater confidence in the results attained.

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Appendix

Table 6: Walking speed distributions used

Parameter	Distribution	Unit	Source
Walking Speed	normal(1.25, 0.21)	ms^{-1}	[86]
Stair Walking Speed (Horizontal)	normal(0.423, 0.130)	ms^{-1}	[87]

Table 7: COVID-19 disease parameters used

Parameter	Value	Unit	Source
Viral Load C_v	lognormal(5.6log10, 2.0log10)	Copies RNA /mL	[25]
Human Infectious Dose, IID	700	Copies RNA	[25]
Volume of Aerosol Expelled Per Breath, V_e	1.92×10^{-6}	MUL/L	[25]
Hand Dose, H_d	3300	Copies RNA	[28]
Hand Area, A_h	uniform(445, 535)	cm^2	[28]
Inhalation Rate, I_r	0.1	L/s	[88]
Exhalation Rate, E_r	0.1	L/s	[88]
Hand To Face Efficiency, η_{HTF}	normal(0.339, 0.132)	(-)	[28]
Mask Efficiency, η_m	uniform(0.75, 0.95)	(-)	[89]
Hand Washing Efficiency,	0.73	(-)	[90]
Surface To Hand Transfer Efficiency, η_{th}	normal(0.118, 0.088)	(-)	[28]
Hand To Surface Transfer Efficiency, η_{ts}	normal(0.123, 0.068)	(-)	[28]
COVID-19 Half-Life in Air	1.1	hours	[28]
COVID-19 Surface Half-Life	5.63	hours	[55]
COVID-19 Skin Half Life	8	hours	[28]